

Program Syllabus:**Semester I**

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCH601	Advanced Database Management System	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To enrich the previous knowledge of database systems and expose the need for distributed database technology.
2. To introduce the fundamental concepts to design and implement aspects of distributed database.
3. To equip the students with the fundamental concepts of object-oriented database, security, and access control.

b. Course Outcomes

CO1	Identify design issues of distributed and object-oriented databases.
CO2	Understand the concepts of DDBMS to summarize the shortcomings of the centralized database system.
CO3	Apply access control mechanisms in distributed databases by granting and revoking privileges.
CO4	Determine the security flaws related to distributed databases
CO5	Create triggers for CRUD operations in object-oriented databases.

c. Syllabus

Module -1	Basics of Data Base Management System
Introduction	An introduction to DBLC, Detail study of phases of DBLC.
DDBMS	Introduction, Types of DDBMS, Levels of Data & Process Distribution (SPSD, MPSD, MPMD)
Module -2	Object Oriented Databases and Data Models
OO Concepts	OO Concepts, O-O Identity, Object Structure and Type Constructors, Encapsulation of Operations, Methods and Persistence, Complex Objects,
Active Database Concepts	Active Database Concepts and Triggers, Temporal Database Concepts, Introduction to Spatial Database Concepts, Introduction to Multimedia Database Concepts
Module -3	Database Security
Database Security Issues	Introduction to Database Security Issues, Discretionary Access Control Based on Granting and Revoking Privileges. Encryption & PKI (Public Key Infrastructure),
Access Control	Comparing Discretionary Access Control and Mandatory Access Control, Role-Based Access Control.
Locking Protocols	Two-phase Locking Protocols, Three Phase Locking Protocol.

d. **Self-study topics for Advance learners:** Degree of Data Abstraction, Transparency Features. Type and Class Hierarchies and Inheritance, Current Trends of Database Technology, Access Control Policies for E-Commerce.

e. Textbooks / Reference Books

1. Elmasri & Navathe, Fundamentals of Database Systems, Fourth Edition.
2. C. J. Date, Introduction to Database Management System.
3. Bipin C. Desai, An Introduction to Database Management Systems, PHI, New Delhi

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
2	24ONMCT60 2	Advanced Computer Networks	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the concepts of data communication
2. To study the functions of different layers used in communication the data over the network.
3. To introduce IEEE standards employed in computer networking.
4. To make the students familiar with different protocols and network components.

b. Course Outcomes

CO1	Identify application layer protocols to support various applications on network.
CO2	Understand the functions of different layers used for communication of the data over the network.
CO3	Apply routing and congestion control algorithms for data transmission.
CO4	Analyse the security issues related to MANET, VANET, and FANET
CO5	Evaluate IEEE standards employed in computer networking

c. Syllabus

Module-1	Introduction to Network
Introduction to computer network	Introduction to computer network, components for communication (mode, medium and media), OSI model, TCP/IP model, physical layer (Digital and Analog Signals, ethernet, IEEE standards)
Network Layer	Network Layer: IP address classes, sub netting, Classless Inter-domain routing (CIDR), ARP, RARP and DHCP concepts, IPv4 & IPv6, The routing protocols: RIP, OSPF, BGP, IP Multicasting, Multicast routing protocols, address assignments, session discovery, etc.
Transport layer	Transport layer: Design issues of transport layer, addressing, establishing connection, flow control and multiplexing, Transport protocols: TCP and UDP.
Module-2	Application Layer and Mobile Computing
Application Layer	WWW, DNS, MIME, HTTP, SMTP, POP, IMAP, FTP, Telnet.
Mobile Computing	Introduction to Mobile Computing, Devices, Networks: Wireline, Wireless, Ad-hoc, Architecture: Architecture of Mobile Computing, 3- Tier Architecture, Presentation (Tier-1), Application (Tier -2), Data (Tier – 3).
Architecture	Comparison of Common wireless system, Architecture of (Wireless Communication)
Module-3	Wireless Communication System
Wireless Communication	Wireless Local Area network (WLAN), Wi-Fi, WiMAX, Wireless Ad-hoc Network, Security issues and challenges in a Wireless network.
Introduction to ad-hoc networks	Introduction to ad-hoc networks – definition, characteristics features, applications Routing Protocols: Design issues, goals and classification, Data Dissemination and Clustering. Advancement is traditional network, MANET, VANET, FANET, Communication in ad-hoc network
SDN	SDN: Introduction and Architecture of Software Defined Network, Characteristics of SDN, Operations, Devices, Controller, Applications of SDN.

d. **Self-study topics for Advance learners:** OSI Model, Mobile Computing, ad-hoc networks.

e. Textbooks / Reference Books

1. Behrouz A Forouzan, “Data Communications and Networking”, McGraw Hill.
2. Andrew S. Tanenbaum, “Computer Networks”, Pearson Education.
3. Subir Kumar Sarkar, T.G. Basavaraju, C. Puttaamadappa, “AdHoc Mobile Wireless Network: Principles, Protocols, and Applications, CRC Press.
4. James F. Kurose, Keith W. Ross, “Computer Networking”, Pearson Education.

5. Michael A. Gallo, William M. Hancock, “Computer Communications and Networking Technologies”, CENGAGE Learning

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
3	24ONMCH603	Web Programming	Prog. Core	4	12
PRE-REQUISITE		Basic Knowledge of web designing/Web Technologies			

a. Course Objectives

1. To design to provide a comprehensive introduction to web techniques.
2. To focus on WWW, HTML, CSS, Java Script & XML.
3. To learn Web Designing and interactive with web pages.

b. Course Outcomes

CO1	Identify the role of PHP as a dynamic webpage creating tool.
CO2	Understand the dynamic web page implementation using PHP
CO3	Implement the CRUD operations in PHP web pages using MySQL.
CO4	Validate form data entered by the user using Ajax validations.
CO5	Create WordPress website using WordPress themes and inbuilt plug-ins.

c. Syllabus

Module-1	Introduction, Array, Forms
Introduction to PHP	Introduction to PHP: PHP for Web Development & Web Applications, History & Future Scope of PHP, Installation of tools for working in PHP like XAMPP, LAMP, WAMP for HP Apache & MySQL Introduction to Language constructs: Variables, constants, Data types, loops, Comments Outputting Data to the Browser: print (), echo (), print_r()
Arrays & Function	Arrays: Introduction to Array, use of array, Numeric Array, Associative Array, Multi-Dimensional Array, converting between arrays and variables, Traversing arrays, Sorting. Function: User Defining functions, Passing parameter & return value Built-in Functions: Math functions, String functions, Array Functions. Date & time functions, Date formats, Include, Require.
PHP Forms & Regular Expressions	PHP Forms: Form Handling, GET, POST, REQUEST, Form Validation, Form Required, Form URL/Email, Form Complete Regular Expressions: Regular Expression Syntax (POSIX), Brackets, Quantifiers, Predefined Character Ranges
Module-2	Objects, File Handling, MySQL, Fetch

Objects, File Handling, MySQL	Objects: Declaring a class, creating an object, accessing properties and methods. PHP File Handling: Understanding file & Directory, File functions, working with directories, building a text editor, File Uploading & Downloading. PHP Session & Cookies: Starting & Destroying PHP Session, turning on auto session, Anatomy of cookie, Setting-accessing-deleting cookies with PHP, Sessions without cookies. PHP MySQL: Connection with MYSQL database, CRUD Operations, setting query parameter, executing query on MYSQL, PHP Joins operations.
Fetch	Fetching Functions: mysqli_query, mysqli_fetch_array, mysqli_fetch_assoc, mysqli_fetch_row, mysqli_fetch_object, mysql_insert_id(). AJAX: Introduction to AJAX, AJAX Model, Implementation of Ajax.
PHP Framework	PHP Framework: Introduction to PHP Framework, Types of Frameworks, Difference Between CMS and MVC.
Module-3	Framework, Themes, WordPress
WordPress	WordPress: Introduction to WordPress, Features, Advantages and Disadvantages of WordPress, Installing WordPress, WordPress Administration Dashboard & Bar, WordPress Settings. Pages and Posts: Difference between Pages and Posts, Creating Posts, Creating Pages, Creating Child Pages Working with Themes: Selecting your Theme, Previewing and Customizing your Theme, Widgets, Using Header and Background Images, Making Other Changes to Themes, Adding CSS with a Child Theme
WordPress Menu	WordPress Menu: Add Menu, Add Menu items, Updating the Menu, editing an Existing Menu Item, adding a Custom Link Menu Item, and Deleting a Menu Item. Working with Plugins: How Plugins Work, Where Plugins Store Their Data, Plugins Setting, Evaluating Plugins, Troubleshooting Plugins. Self-Study: Child Pages
Web Deployment	Web Deployment: Managing Domains & Hosting, Working with CPanel, Paypal Payment Gateway Integration,

d. Textbooks / Reference Books

1. Steven Holzner, "PHP-The complete reference", Mc GrawHill(2007).
2. Robin Nixon, "PHP-MYSql-JavaScript", O'Reilly (2009).
3. Rasmus Lerdorf, Kevin Tatroe, Bob Kaehms, Ric McGredy, 2002, Programming PHP, O'REILLY (SPD).

g. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
4	24ONMCH604	Python Programming	Prog. Core	4	12
PRE-REQUISITE		--			

a. Course Objectives

1. To develop Python programs with conditionals and loops.
2. To develop Python Graphical User Interface and to develop Python programs that can read and write data from/to files.
3. To store and retrieve data from database using Python program.

b. Course Outcomes

CO1	Identify the core construct of python language and its data structure best suited for problem solving.
CO2	Understand the use of module, function, in-built string functions in problem solving.
CO3	Implement the GUI based application using the concept of classes and object
CO4	Create GUI based application using database for handling operations of the organization.

c. Syllabus:

Module-1	Introduction to Python and Basics
Introduction to python	Introduction to python, features, setting up programming environment, python program structure, Tokens: Keywords, identifier, variables, data types, operators.
Python Sequence	Lists-Accessing elements, Index position, Using individual values from a list, Changing/Adding and removing Elements, Organizing a list, Loop through an entire list, Avoiding Indentation Error, Numerical Lists, Sublist, List comprehension, Tuple, Set, Dictionary- Working with dictionaries, Looping through dictionaries, Nesting.
Experiment no. 1	1. Write a program to print twin primes less than 1000. If two consecutive odd numbers are both prime, then they are known as twin primes. 2. Write a program to implement these formulae of permutations and combinations. Number of permutations of n objects taken r at a time: $p(n, r) = n! / (n-r)!$ Number of combinations of n objects taken r at a time is: $c(n, r) = n! / (r! (n-r)!) = p(n, r) / r!$
Conditional Statements	Concept of indentation If, If Else, Switch; Looping: For, While, Nested loops, Jumping statements
Experiment no. 2	Two different numbers are called amicable numbers if the sum of the proper divisors of each is equal to the other number. For example, 220 and 284 are amicable numbers. Sum of proper divisors of 220 = $1+2+4+5+10+11+20+22+44+55+110 = 284$ Sum of proper divisors of 284 = $1+2+4+71+142 = 220$ Write a function to print pairs of amicable numbers in a range.
Function	Passing arguments, Return Values, Passing a list, Passing an Arbitrary number of arguments, Function in Modules, Recursive function, Nested functions, Default and flexible arguments, Lambda function, Map() function, Filter function, Reduce() function, Python inbuilt functions
Experiment no. 3	WAP to get user id, user name, and user age from user and based on the entered id print the details for particular user. Hint: use dictionary.
Class	Creating and using a class, working with classes and Instances, public and private members, Inheritance, types of inheritance, Polymorphism, Importing Classes, Python Standard library.
Experiment no. 4	1. Implement a Student class with information such as rollno, name, class. The information must be entered by the user. Create a program to implement library management system using classes and objects
Module-2	Data Analytics
Introduction to Data Analytics	Introduction to Data Analytics, requirements of data analytics in Python. Introduction to Numpy, features, environment setup, numpy Ndarray, array creation data types, array attributes, numpy operations, mathematical and statistical functions
Experiment no. 5	1. Write an experiment to swap two columns in numpy array 2. Write an experiment import a dataset with numbers and texts keeping the text intact in python numpy?
Introduction to matplotlib	Introduction to matplotlib, Figure class, Axes class, line plot, subplots, Bar plot, histogram, scatter plot, pie chart, box plot, area chart, word cloud, Bee swarm plot, violin graph, Working with text

Experiment no. 6	1. Write a python program to generate a simple bar graph using matplotlib. The graph should be properly labeled. 2. Write a python program to generate Pie-chart using matplotlib. The graph should be properly labeled. 3. Write a Python program to plot the function $y = x^2$ using the matplotlib libraries.
Experiment no. 7	1. Write a program in Python to compute the greatest common divisor and the least common multiple of two integers. 2. Write a program in Python to test if a number is equal to the sum of the cubes of its digits. Find the smallest and largest such numbers in the range of 100 to 1000
Experiment no. 8	1. Write a program in python to read sort a list of integer elements using the bubble sort method. Display the sorted element on the screen. 2. Write a program in python to find out the frequency of each element in a list using a dictionary.
SELF STUDY TOPIC	numpy matrix, 3D graph
Module-3	Statistics
Statistics	Central tendency, measure of dispersion, correlation and regression. Introduction to pandas, data structure, implement statistics. Handling data frame, read csv, xls files. Handling null values or missing data, group by. Data visualization with different existing dataset. Random walks, PDF.
Experiment no. 9	Perform different statistics operations on dataset taken from kaggle or datagov
Data visualization using seaborn and plotly	Introduction to seaborn, relational plots, categorical plots, distribution plots, matrix plots. Plotly: Line, bar, scatter, histogram, violin, gantt, heatmap, 3D graph.
Experiment no. 10	Visualize dataset using plotly and create heatmap of the correlation between different columns
SELF STUDY TOPIC	correlation matrix

d. Self-study topics for Advance learners: Dictionary comprehension, numpy matrix, 3D graph, correlation matrix.

e. Textbooks / Reference Books

1. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2nd edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016
(<http://greenteapress.com/wp/thinkpython/>)
2. Michael Urban, Joel Murach, Mike Murach: Murach's Python Programming; Dec, 2016.
3. Guido van Rossum and Fred L. Drake Jr, An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2011.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
5	24ONMCH605	Network Security and Cryptography	Prog. Core	4	12
PRE-REQUISITE		--			

a. Course Objectives

1. To introduce various encryption and authentication techniques for network security
2. To obtain knowledge on standard algorithms used to provide confidentiality, authenticity, and Integrity
3. To secure a message over the insecure channel by various means.

b. Course Outcomes

CO1	Identify standard algorithms to provide confidentiality, authentication and integrity of the data over the networks
CO2	Understand Security services and policies to provide a secure network.
CO3	Classify Cryptographic techniques for network security.
CO4	Implement cryptographic techniques for message passing to secured network
CO5	Evaluate the performance of the network using Firewall and packet filtering techniques

c. Syllabus

Module-1	Introduction to Network Security
Introduction to Security	Introduction to Security: Need for security, Security approaches, Policies of security, Types of attacks, Services: confidentiality, integrity, availability.
Encryption Techniques	Encryption Techniques: Plaintext, Cipher text, Substitution & Transposition techniques, Encryption & Decryption, Cryptographic attacks, Key range & Size. Symmetric & Asymmetric Key Cryptography: Algorithm types & Modes, DES, IDEA, Differential & Linear Cryptanalysis, Symmetric & Asymmetric key together.
Authentication	Authentication basics, Passwords, Authentication tokens, Certificate based & Biometric authentication.
SELF STUDY TOPIC	Knapsack algorithm
Module-2	Authentication
Cryptography	Cryptography: Secure inter branch payment transactions, Conventional Encryption and Message Confidentiality, Conventional Encryption Principles, Conventional Encryption Algorithms
Key Distribution & Management	Key Distribution & Management: KDC, Kerberos and certificate authorities
Public Key Cryptography	Public Key Cryptography and Message Authentication: Approaches to Message Authentication, handshake mechanism, Hash function, SHA-1, MD4, MD5, Public-Key Cryptography Principles, RSA, Digital Signatures.
SELF STUDY TOPIC	Location of Encryption Devices
Module -3	Firewalls and Web Security
Firewalls	Packet filters, Application-level gateways, Encrypted tunnels, Cookies, Web security problems
Email Security	Distribution lists, Establishing keys, Privacy, source authentication, message integrity, non-repudiation, proof of submission, proof of delivery, message flow confidentiality, anonymity, Pretty Good Privacy (PGP).
SELF STUDY TOPIC	Viruses and malware

d. Self-study topics for Advance learners:Knapsack algorithm, Location of Encryption Devices, Viruses and malware

e. Textbooks / Reference Books

1. Douglas Stinson, "Cryptography Theory and Practice", 2 nd Edition, Chapman & Hall/CRC.
2. B. A. Forouzan, "Cryptography & Network Security", Tata Mc Graw Hill.
3. W. Stallings, "Cryptography and Network Security", Pearson Education.
4. Kaufman, c., Perlman, R., and Speciner, M., Network Security, Private Communication in a public world, 2nd ed., Prentice Hall PTR., 2002.
5. Cryptography and Network Security; McGraw Hill; Behrouz A Forouzan.
6. Information Security Intelligence Cryptographic Principles and App. Calabrese Thomson

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Semester 2

S. N	Course Code	Course Title	Course Type	Credit	Week
6	24ONMCH651	Advanced Internet Programming	Prog. Core	4	12
PRE-REQUISITE		--			

a. Course Objectives

1. To give a strong foundation on Advanced Java Programming Techniques.
2. To write a computer program to solve specified problems.
3. To understand concepts of programming such as conditional and iterative execution variables, methods, etc.

b. Course Outcomes

CO1	Understand the basic concepts of Static vs Dynamic Web, enterprise applications and client-server architecture
CO2	Create server-side web applications using servlet and JSP.
CO3	Implement database connectivity to perform CRUD commands in server-side applications.
CO4	Analyse Hibernate architecture to interact with the database using Java Objects
CO5	Create server-side applications using NodeJS.

c. Syllabus

Module-1	Client Server architecture and Servlet
Introduction of web Applications	Types of web Applications, Web Application, Web Page, Websites, Type of Websites, static and Dynamic page.
Client-side Application	Client-side Application, Server side Application. Client Server architecture, Introduction of Java 2 Enterprise Edition.
Server-Side Programming	Server-Side Programming, Web Server, Java Server side components, Servlet Architecture, Web Container, Servlet Life Cycle, Tomcat Interface, Servlet interface
Servlet	Types of Servlet, HttpServletRequest and HttpServletResponse, GET and POST request methods, Retrieving data from database to Servlet, Servlet Collaboration: Request Dispatcher and send Redirect, ServletConfig and ServletContext
Module-2	JSP and JDBC
Introduction to JSP	Introduction to JSP and its advantages over Servlet, Architecture of JSP, Elements of JSP, Scripting elements, Directives and actions, Custom Tags, JSP configuration, implicit objects.
Java Data Base Connectivity (JDBC) using Servlet and JSP	JDBC and ODBC, Types of Drivers for connection, CRUD operations, Statement and Prepared Statement interface, query and execute Query, Result Set interface, Result Set Meta Data interface. CRUD. Applications using Servlet and JSP
Module-3	Hibernate and NodeJS
Database Operations	Introduction to Hibernate, Architecture of Hibernate Database Operations: Insert, Update, Delete, Select
Node JS	Node JS: Introduction to JavaScript. Creating functions, Introduction to Node JS, npm, V8, Asynchronous vs Synchronous request, REPL, Reading and writing into file and directories, working with buffer and stream, Creating Server with Http request, Event, Process, Web Modules, Cryptography in NodeJS, NodeJS CRUD.

d. Self-study topics for Advance learners: Web Application Servers, Use Bean Tag, Form Validation

e. Textbooks / Reference Books

1. Herbert Schildt, “Java: The Complete Reference”, Tenth Edition, McGraw-Hill Education, 2017
2. Gavin King, Christian Bauer, “Java Persistence with Hibernate”, Manning publisher, New York-USA.
3. Y. D. Liang,” Introduction to Java Programming”, Pearson Education.
4. JAVA 2 Unleashed, Tech Media Publications, New Delhi.
5. JAVA 2(1.3) API Documentations.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
7	24ONMCH65 2	Design and Analysis of Algorithms	Prog. Core	4	12
PRE-REQUISITE		--			

a. Course Objectives

1. To demonstrate familiarity with major algorithms and data structures.
2. To analyze the performance of algorithms.
3. To apply important algorithmic paradigms and methods of Analysis.

b. Course Outcomes

CO1	Analyze the asymptotic performance of algorithms.
CO2	Implement major data structure algorithms.
CO3	Apply and analyze important algorithmic design paradigms and their applications.
CO4	Implement the major graph algorithms to model engineering problems.
CO5	Synthesize efficient algorithms in common engineering design situations.

c. Syllabus

Module-1	Introduction
Introduction	Characteristics of algorithm, Algorithm Specification, Analysis Framework
Performance Analysis	Space complexity, Time complexity
Asymptotic Notations	Big-Oh notation(O), Omega notation(Ω), Theta notation(Θ), and Little-oh notation(o), Mathematical analysis of Non-Recursive and recursive Algorithms With Examples.
Important Problem Types	Sorting, Searching, String processing, Graph Problems, Combinatorial Problems.
Fundamental Data Structures	Linked lists, Stacks, Queues, Graphs, Trees, AVL tree, B Trees, Sets
Module-2	Algorithm Design Paradigm
Divide and conquer	General method, Binary search, Merge sort, Quick sort, Advantages and disadvantages of divide and conquer. Decrease and Conquer approach: Topological Sort.
Greedy Method	General method, Coinchange Problem, Fractional Knapsack Problem
Minimum cost Spanning trees	Prim's Algorithm, Kruskal's Algorithm, Single source
Shortest paths	Dijkstra's Algorithm
Optimal Tree Problem	Huffman Trees and Codes
Transform and Conquer Approach	Heap sand Heap Sort
Module-3	Dynamic Programming
Dynamic Programming	General method with Examples, Multistage Graphs
Transitive Closure	Warshall's Algorithm, All Pairs
Shortest Paths	Floyd's Algorithm, Optimal Binary Search Trees, Knapsack Problem 0/1, Bellman-Ford Algorithm, Travelling Sales Person problem, Reliability design.
Backtracking: General method	N-Queens problem, Sum of subset problem, Graph Coloring. Hamilton cycles.
Branch and Bound	Assignment Problem, Travelling Sales Person problem, Knapsack 0/1.
Knapsack Problem	LC Branch and Bound solution

d. Self-study topics for Advance learners: Dictionaries, Job sequencing with Deadlines, FIFO Branch and Bound solution.

e. Textbooks/Reference books

1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", 2nd Edition, 2009. Pearson.
2. Satraj Sahni and Rajasekaran, "Computer Algorithms/C++", Computer Science Press, 1997

3. Ellis Horowitz”, “fundamentals of computer algorithms “, 2nd Edition, 2014, Universities Press
4. Thomas H.Cormen, Charles E.Leiserson, Ronal L. Rivest, “Introduction to Algorithms”, Clifford Stein, 3rdEdition, PHI
5. S.Sridhar, “Design and Analysis of Algorithms”, Oxford(Higher Education).

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
8	24ONMCT653	Software Testing	Prog. Core	4	12
PRE-REQUISITE		--			

a. Course Objectives

1. To test software in structured, organized ways.
2. To design quality tests effectively.
3. To implement testing strategies to real-world applications

b. Course Outcomes

CO1	Comprehend the basics of software development life cycle and software testing
CO2	Implement the test strategies using JIRA software
CO3	Analyze the relationship between software modules during integration testing.
CO4	Monitor test progress of healthcare applications using JIRA Software.
CO5	Design test cases to find software bugs.

c. Syllabus

Module-1	Fundamentals of Testing
Software Development Life Cycle	Software Development Life Cycle (SDLC), SDLC Models (Waterfall Model, V Model, Agile Model, Rapid Application Development), Impact of software bugs, Objective of testing, Testing principles
Experiment 1	Testing principles through illustrations with respect to different SDLC Models
Software Testing Life Cycle	Software Testing Life Cycle, establishing test policy, test factors and eleven steps of software testing process, Testing documentation using IEEE829, Test plan and Test Report, Test Metrics, Traceability Matrix
Experiment 2	Create a Test plan for online websites like Rediff, LinkedIn etc.
Test Levels	Roles & Responsibilities of Quality Assurance Engineer, Test Levels (Unit, Component, Module, Integration, System, Acceptance, Generic), Software testing pyramid
Experiment 3	Design test cases for register new user page of classic website like Rediff, LinkedIn for every field with accepted values (valid values)
Module-2	Different approaches to Testing
Static Testing	Static Testing: Structured Group Examinations, Static Analysis, Control flow & Data flow, Determining Metrics
Experiment 4	Design test cases for register new user page of classic website like Rediff, LinkedIn for every field with unaccepted values (invalid values) and verify the warning and error
Dynamic Testing	Dynamic Testing: Black Box Testing (Equivalence Class Partitioning, Boundary Value Analysis, Cause Effect Graphing and Decision Table Technique)
Experiment 5	Design test cases for register new user page of classic website like Rediff, LinkedIn for every field with boundary values (minimum and max values for the fields)
Experiment 6	Design test cases using decision table for login page of rediff and linkedin page
White Box Testing	White Box Testing (Statement Coverage, Branch Coverage, Test of Conditions, Path Coverage), Gray Box Testing, Intuitive and Experience, Based Testing, Alpha, Beta, Performance, Load and Stress Testing, Key Performance Indicator (KPI's) of software testing
Experiment 7	Live Testing Projects: Testing application of different domains: (Ecommerce, Educational, Travel, Healthcare, etc.)
SELF STUDY TOPIC	Exploratory testing and Planned testing
Module-3	Test Management using JIRA
Introduction To JIRA	Introduction To JIRA, Test Management In JIRA, Advanced Search and Introduction to JQL (JIRA Query Language), different types of issues in JIRA (sub-task, bug, epic, improvement, new feature, story, task), Jira Dashboards, Different methods for creating issue
Experiment 8	Create different types of issues (sub-task, epic, improvement, new feature, story, task) using different methods
Experiment 9	Convert issue to Subtask and Vice versa, edit. email, label and move an issue
Experiment 10	Jira Dashboards, Search Tasks using various search features provided by JIRA, log work hours and set label in JIRA

Defect and Bug	Difference between Defect and Bug, Defect Life Cycle, Defect Tacking Tools, create a Bug Report, Severity & Priority
Experiment 11	Bug management in JIRA (Create, track, resolve, close and reopen)
Different types reports	Work Flows, plug-ins in JIRA, Use of Clone and Link in JIRA, Export and import data in Jira with different formats, Different types reports (Agile, Issue Analysis, Forecast & Management, etc.),
Experiment 12	Clone and link an issue, import export data, generate different types of reports
SELF STUDY TOPIC	JIRA Agile

d. Self-study topics for Advance learners:How to establish testing policy, ISO and IEEE standards for quality of products models, Exploratory testing and Planned testing, JIRA Agile,

e. Textbooks / Reference Books

1. Software Testing by Ron Patton, Sams
2. Software Quality Assurance, by Daniel Galin, Pearson Education
3. Foundations of Software Testing by Aditya P Mathur, Pearson Education
4. Testing and Quality Assurance for Component-based Software, by Gao, Tsao and Wu Artech House Publishers.
5. Handbook of Software Quality Assurance, by G. Gordon Schulmeyer, James I. McManus, Second Edition, International Thomson Computer Press
6. Software Quality, by Mordechai Ben-Menachem/Garry S. Marliss, by Thomson Learning publication.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
9	24ONMCH654	Web Application Development	Prog. Elective	4	12
PRE-REQUISITE					

a. Course Objectives

1. To acquire knowledge on the usage of the .Net platform in developing web applications.
2. To develop skills in analyzing the usability of a website.
3. To understand how to plan and conduct user research related to web usability

b. Course Outcomes

CO1	Understand the .net framework and its various components used to design web applications
CO2	Implement server-side applications using web controls.
CO3	Analyze the concepts of C# programming, including Indexer, Generics, Properties, Delegates.
CO4	Design WPF applications on visual studio.
CO5	Create classes of the System.xml namespace to experiment with XML file operations.

c. Syllabus

Module-1	Overview of .Net Framework
.NET Framework Introduction	Overview, Framework Components ,Framework Versions , Types of Applications which can be developed using MS.NET ,MS.NET Base Class Library,MS.NET Namespaces ,MSIL / Metadata and PE files, The Common Language Runtime (CLR), Managed Code ,MS.NET Memory Management / Garbage Collection, Common Type System (CTS) ,Common Language Specification (CLS) ,Types of JIT Compilers, Security Manager. VS.NET and C, Introduction to Project and Solution in Studio, Command Line Arguments, Global, Stack and Heap Memory, Reference Type and Value Type, Boxing and Unboxing, Pass by value and by reference and out parameter, Array Lists & Hash Tables, Generic Collections.
Experiment 1	WAP to Demonstrate Constructor Calling
Experiment 2	WAP to Demonstrate Array Implementation.
.Net Assembly	Classification of Assembly, Creating and using Managed DLLs, Private Assembly and Shared Assembly, The Global Assembly Cache, Property. Procedures. File Handling: System.IO Namespace, working with Directories and Files, Read and write file, Stream Reader and Stream Writer Classes.
Experiment 3	WAP to Demonstrate Properties. WAP to Demonstrate Generics.
Experiment 4	WAP to Demonstrate Indexers.
Module-2	Configuring Controls in Windows Forms
Windows Forms and Controls and MDI Applications	Control Class, Buttons, Text Boxes, Labels, Literals, Image Controls, Picture Box Control, Panel Control, Combo Box Control, List Boxes, Dropdown Lists, Date Time Picker Control, Link Labels, Check Boxes, Check Box Lists, Radio Buttons, Radio Button Lists, Rich Text Box Control, Tab Control, Tool Strip Control, Menu Strip Control, Progress Bar, MDI Applications, MDI Parent andMDI Child Forms, Manage Menus
Experiment 5	WAP to Demonstrate the use of controls.
Configuring Connections and Connecting to Data	What Is a Connection Object, Creating Connections in Server Explorer, Creating Connections Using Data Wizards, Creating Connection Objects Programmatically? Opening and Closing Data Connections. Working with Data in a Connected Environment: Creating and Executing CommandObjects, working with Parameters in SQL Commands.
Configuring Connections and Connecting to Data	What Is a Connection Object, Creating Connections in Server Explorer, Creating Connections Using Data Wizards, Creating Connection Objects Programmatically? Opening and Closing Data Connections. Working with Data in a Connected Environment: Creating and Executing CommandObjects, working with Parameters in SQL Commands.
Experiment 6	WAP to Demonstrate Command Object. WAP to Demonstrate Data Adapter Object.
Experiment 7	WAP to create the connection using different ways.
Module-3	Configuring Connections and Connecting to Data
Create, Add, Delete, and Edit Data in a Disconnected Environment	Creating Data Set Objects, Creating Data Table Objects, Creating Data Adapter Objects, Working with Data in Data Table Objects.
Experiment 8	WAP to Demonstrate DataSet and DataTable class.

N-Tier Layered Architecture Application	Understanding Tier and Layer, Dividing, Application into multiple layers.XML: Reading and Writing XML, Important Classes in the System. XML, Namespace, Read and Write XML Nodes and Attributes.
Experiment 9	WAP to add data in xml file. WAP to write data into xml file.
Building Setup Applications	Installation Package, customize a Setup Project, Control Installation of an Application, Specify Conditions of an Install, Custom Actions for after anInstallation.
Experiment 10	WAP to Create the setup of windows application.

d. Self-study topics for Advance learners: Frame Work, Assembly, Controls, Layers, Package

e. TextBooks and Reference Books

T1 Matthew MacDonald, “Beginning ASP.NET 3.5 in C# 2008” by a Press.

T2 Beginning.NET with C# (Wrox Beginning Guides

R1 Stephen Walther, “ASP.NET 3.5 UNLEASHED” by published by Sams.

R2 .NET 4.5 by Kogent Learning Solutions Inc.(Author) Black Book

g. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
1 2	24ONMCT655	Cyber Security	Prog. Elective	4	12
PRE-REQUISITE					

a. Course Objectives

1. To Create cyber security awareness and to understand principles of web security
2. To understand key terms and concepts in cyber law, intellectual property and cyber crimes, trademarks and domain theft.
3. To make attentive to students about possible hacking and threats in this communication era.

b. Course Outcomes

CO1	Identify the utilization of security services.
CO2	Understand the Intellectual property rights and Open Standards.
CO3	Apply the Protection and resilience for Critical Information Infrastructure.
CO4	Analyse the issues for creating Security Policy for a Large Organization.
CO5	Evaluate website using IT Act and Criminal Procedural Code.

c. Syllabus

Module-1	Cyber Security Fundamentals
Security Concepts	Authentication, Authorization, Non-repudiation, Confidentiality, Integrity, availability. Cyber Crimes and Criminals: Definition of cyber-crime, types of cyber-crimes and types of cyber-criminals.
Chapter 1.2	Anti-forensics: Use of proxies, use of tunnelling techniques. Fraud techniques: Phishing and malicious mobile code, Rogue antivirus, Click fraud. Threat Infrastructure: Botnets, Fast Flux
SELF STUDY TOPIC	advanced fast flux.
Module-2	Information Technology Act 2000
Techniques to gain foothold	Shellcode, Buffer overflows, SQL Injection, Race Conditions, DoS Conditions, Brute force and dictionary attacks. Misdirection, Reconnaissance, and Disruption Methods: Cross-Site Scripting (XSS), Social Engineering, WarXing.
Cyber Acts and Regulations	Overview of IT Act 2000, Amendments and Limitations of IT Act, Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature, Certifying Authorities, Cyber Crime and Offenses, Network Service Providers Liability, Cyber Regulations Appellate Tribunal, Penalties and Adjudication.
SELF STUDY TOPIC	DNS Amplification Attacks
Module-3	Cyber Law and Related Legislation
Chapter 3.1	Patent Law, Trademark Law, Copyright, Software Copyright or Patented, Domain Names and Copyright disputes, Electronic Data Base and its Protection, IT Act and Civil Procedure Code.
Chapter 3.2	IT Act and Criminal Procedural Code, Relevant Sections of Indian Evidence Act, Relevant Sections of Bankers Book Evidence Act, Relevant Sections of Indian Penal Code, Relevant Sections of Reserve Bank of India Act, Law Relating To Employees And Internet, Alternative Dispute Resolution
SELF STUDY TOPIC	Online Dispute Resolution (ODR).

d. Self-study topics for Advance learners:advanced fast flux, DNS Amplification Attacks, Online Dispute Resolution (ODR).

e. Textbooks / Reference Books

1. Gerald R. Ferrera, Margo E. K. Reder, "Cyber Law Text & Cases", CENGAGE LEARNING Publication.
2. Georgia Weidman, Penetration testing A Hands-On Introduction to Hacking, no starch press, 2014.
3. Charles P. Pfleeger Shari Lawrence Pfleeger Jonathan Margulies, Security in Computing, 5th Edition Pearson Education, 2015
4. James Graham et al., "Cyber Security Essentials", CRC Press.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus: Semester III - Cloud Computing

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT701	Introduction to Cloud Computing	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To provide an in-depth understanding of cloud computing concepts, virtualization technologies, and modern cloud architectures.
2. To equip students with knowledge about different types of cloud computing environments and virtualization approaches.
3. To develop skills in implementing and managing cloud-based technologies effectively.

b. Course Outcomes

CO1	Understand and define the fundamental concepts of cloud computing.
CO2	Identify different types of cloud environments and virtualization techniques.
CO3	Evaluate and compare various cloud computing and virtualization technologies.
CO4	Apply cloud computing solutions to solve business challenges.
CO5	Analyze the impact of cloud technologies on business and IT strategy.

c. Syllabus

Module -1	Introduction to Cloud Computing
Chapter 1:	Cloud Computing Fundamentals - Introduction to cloud computing - Definition and history of cloud computing - Roots of cloud computing - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring the evolution and impact of cloud technologies.
Chapter 2:	Features of Clouds - Introduction to features of cloud computing - Detailed exploration of cloud features - Summary and self-assessment questions - SELF STUDY TOPIC: Deep dive into the scalability and elasticity of cloud computing.
Chapter 3:	Types of Cloud Computing - Classification of cloud models: Public, Private, Hybrid, and Community - Summary and self-assessment questions - SELF STUDY TOPIC: Comparing security features across different cloud models.
Chapter 4:	Properties and Characteristics of Cloud Computing - Discussion on properties, characteristics, and disadvantages of cloud computing - Comparison of cloud computing with cluster and grid computing - Summary and self-assessment questions - SELF STUDY TOPIC: Analysis of cloud computing cost-effectiveness and operational impacts.
Module -2	Virtualization in Cloud Computing
Chapter 5:	Virtualization - Introduction and characteristics of virtual environments - Taxonomy of virtualization techniques - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced virtualization configurations and optimizations.
Chapter 6:	Virtualization and Cloud Computing - Implementation levels of virtualization - Pros and cons of virtualization - Summary and self-assessment questions - SELF STUDY TOPIC: The role of virtualization in disaster recovery scenarios.

Chapter 7:	Types of Virtualization - Overview of different types of virtualization: Data, Hardware, Software, Storage, Server, OS - Focus on Linux and Windows virtual environments - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring the impact of OS-level virtualization on system performance.
Chapter 8:	Virtualization Technology - Structure of virtualization and virtual machine technology - Applications and pitfalls of virtualization in enterprises - Summary and self-assessment questions - SELF STUDY TOPIC: Virtualization security risks and mitigation strategies.
Module -3	Advanced Cloud Computing Architectures
Chapter 9:	Cloud Computing Architecture - Fundamental concepts, models, roles, and boundaries in cloud architecture - Summary and self-assessment questions - SELF STUDY TOPIC: The future of cloud architectures and emerging trends.
Chapter 10:	Characteristics of Cloud Computing Architecture - Cloud service models and the economics of cloud computing - Open challenges in cloud architecture - Summary and self-assessment questions - SELF STUDY TOPIC: Cloud service model innovations and market impacts.
Chapter 11:	Service Oriented Architecture and the Cloud - Defining and understanding Service Oriented Architecture (SOA) - Coupling within cloud services - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced SOA design patterns and best practices.
Chapter 12:	Implementation of SOA - Services in the cloud and their business applications - Integrating SOA with cloud computing strategies - Summary and self-assessment questions - SELF STUDY TOPIC: Case studies on successful SOA implementations in large enterprises.
Chapter 13:	Industrial Platforms and New Developments - Overview of major cloud platforms: Amazon Web Services, Google App Engine, Microsoft Azure - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced uses of cloud platforms in AI and machine learning applications.

d. Self-study topics for Advance learners:

- Evolution and impact of cloud technologies
- Scalability and elasticity of cloud computing
- Security features across different cloud models
- Cloud computing cost-effectiveness and operational impacts
- Advanced virtualization configurations and optimizations
- Role of virtualization in disaster recovery scenarios
- Impact of OS-level virtualization on system performance
- Virtualization security risks and
- Mitigation strategies
- Future of cloud architectures and emerging trends
- Cloud service model innovations and market impacts
- Advanced SOA design patterns and best practices
- Case studies on successful SOA implementations in large enterprises
- Advanced uses of cloud platforms in AI and machine learning applications

e. Textbooks / Reference Books

1. Cloud Computing: Concepts, Technology & Architecture, Thomas Erl, Ricardo Puttini, and Zaigham Mahmood.
2. Mastering Virtualization, Timothy Warner.
3. Cloud Architecture Patterns, Bill Wilder.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT702	Introduction to Amazon Web Services	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To provide an in-depth understanding of Amazon Web Services (AWS) and its key components.
2. To equip students with knowledge on building and managing virtual infrastructures using AWS.
3. To develop skills in securing systems, storing data, and automating operations within the AWS ecosystem.

b. Course Outcomes

CO1	Understand the fundamental concepts and services of AWS.
CO2	Build and manage virtual infrastructure using AWS technologies.
CO3	Implement security measures and manage data storage and sharing solutions on AWS.
CO4	Automate and optimize AWS operations using advanced AWS tools and services.
CO5	Analyze the impact of AWS solutions on business and IT strategies.

c. Syllabus

Module -1	Introduction to Amazon Web Services (AWS)
Chapter 1:	What is Amazon Web Service - What is cloud computing - Working with AWS - Benefits of AWS - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring case studies of successful AWS implementations.
Chapter 2:	Exploring AWS Services - AWS service exploration - Interacting with AWS - Creating AWS - Summary and self-assessment questions - SELF STUDY TOPIC: Deep dive into AWS's most popular services and their use cases.
Introduction to Microsoft Azure services	Billing Alarm of AWS - AWS Bills - Track of AWS bills - Summary and self-assessment questions - SELF STUDY TOPIC: Strategies for optimizing AWS spending.
Chapter 4:	WordPress - Infrastructure creator - Exploring infrastructure - Costing - Deletion - Summary and self-assessment questions - SELF STUDY TOPIC: Managing WordPress on AWS from setup to scale.
Module -2	Building Virtual Infrastructure
Chapter 5:	Using Virtual Machines - Exploration of virtual machines - Monitoring virtual machines - Allocation of virtual machine - Summary and self-assessment questions - SELF STUDY TOPIC: Best practices in VM performance optimization.
Chapter 6:	Programming Your Infrastructure - Infrastructure as code - Using interface - Programming with SDK - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced techniques in infrastructure as code.

Chapter 7:	Automating Deployment - Deploying applications - Comparing development tools - Creating and running virtual machines - Summary and self-assessment questions - SELF STUDY TOPIC: Continuous integration and continuous deployment (CI/CD) with AWS.
Module -3	System Serving and Data Storing and Sharing
Chapter 8:	Securing Your System - Who is responsible for security - Updated software - Securing AWS - Controlling traffic - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced security measures and compliance on AWS.
Chapter 9:	Automating Operational Tasks - Code with AWS Lambda - Health check with AWS Lambda - Working with AWS Lambda - Summary and self-assessment questions - SELF STUDY TOPIC: Using AWS Lambda for serverless computing.
Chapter 10:	Storing the Objects - What is an object store - Amazon S3 - Backing up the data - Archiving and storing data - Summary and self-assessment questions - SELF STUDY TOPIC: Data lifecycle management with Amazon S3.
Chapter 11:	Storing Data on Hard Drive - Elastic Block Store (EBS) - Instance or temporary store - Summary and self-assessment questions - SELF STUDY TOPIC: Comparing EBS and other AWS storage options.
Module -4	Sharing Data
Chapter 12:	Sharing Data Volumes - File system creation - Mount target - Sharing files as EC2 - Summary and self-assessment questions - SELF STUDY TOPIC: Building scalable file systems with Amazon EFS.

Chapter 13:	Relational Database - Starting MySQL database - Data importation - Data restoration and backing up - Controlling data access - Summary and self-assessment questions - SELF STUDY TOPIC: Managing relational databases on AWS.
Chapter 14:	Caching Data - Cache cluster creation - Cache deployment options - Cache access - Summary and self-assessment questions - SELF STUDY TOPIC: Optimizing application performance with Amazon ElastiCache.

d. Self-study topics for Advance learners:

- Case studies of successful AWS implementations.
- Deep dive into AWS's most popular services and their use cases.
- Strategies for optimizing AWS spending.
- Managing WordPress on AWS from setup to scale.
- Best practices in VM performance optimization.
- Advanced techniques in infrastructure as code.
- Continuous integration and continuous deployment (CI/CD) with AWS.
- Advanced security measures and compliance on AWS.
- Using AWS Lambda for serverless computing.
- Data lifecycle management with Amazon S3.
- Comparing EBS and other AWS storage options.
- Building scalable file systems with Amazon EFS.
- Managing relational databases on AWS.
- Optimizing application performance with Amazon ElastiCache.

e. Textbooks / Reference Books

1. AWS Certified Solutions Architect Study Guide, by Ben Piper and David Clinton.
2. Amazon Web Services in Action, by Andreas Wittig and Michael Wittig.
3. AWS for Developers For Dummies, by John Paul Mueller.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT703	Introduction to Microsoft Azure services	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To provide a comprehensive understanding of Microsoft Azure's infrastructure and services.
2. To equip students with the skills necessary to design, deploy, and manage applications on Azure.
3. To foster a deep understanding of cloud architecture principles using Azure as a platform.

b. Course Outcomes

CO1	Understand and articulate the core components and services of Microsoft Azure.
CO2	Design and implement Azure virtual networks and manage associated resources effectively.
CO3	Deploy, manage, and secure virtual machines and web applications on Azure.
CO4	Utilize Azure SQL Database and Azure Storage solutions for data management.
CO5	Apply knowledge of Azure services to real-world cloud solutions and case studies.

c. Syllabus

Module -1	Getting Started with Microsoft Azure
Chapter 1	Getting Started with Microsoft Azure - Introduction to Azure, azure architecture - Why, What and Benefits of Azure - Azure Hosting Models - Azure Services - Azure Resource Group - Installing Microsoft Azure SDK - Summary and self-assessment questions - SELF STUDY TOPIC: Explore Azure architecture and its impact on cloud solutions.
Chapter 2	Fundamentals of Azure Virtual Networks - Overview of Azure Networking - Benefits of Virtual Network - Understanding Network Resources - Create a VNet using Azure, Setup Network Security Group - Create a Public IP Address - Create Network Interface Card with public, and private IP - Create a Virtual Machine - Azure Application Gateway, Understanding Azure DNS - Design and implement a multi-site or hybrid network - Virtual Clusters and Resource management - Summary and self-assessment questions - SELF STUDY TOPIC: Implementing hybrid connectivity with Azure.
Chapter 3	Azure Virtual Machines - Introduction - About Virtual Machine Workloads - Deploy popular application frameworks using Azure Resource Manager templates - Understand and Capture VM Images, Upload an on-premise VHD to Storage Account, Deploy a New VM from the Captured Image - Virtual Machine Disk Types and VM Storage, Virtual Machine Sizes in Azure - Azure VM Backup and Restore Services, different series of virtual machines - Summary and self-assessment questions - SELF STUDY TOPIC: Management and scalability of Azure VMs
Chapter 4	Web Apps and Services by Azure - Introduction to App Service - Application Types, Deploy Web Apps - Deploying Web App directly from Visual Studio - App Service plans; Create App Service Plan, Migrate Web Apps between App Service plans, Create a Web App within an App Service plan - Configuring Web Apps, Application Settings Configuration, Database Connection Strings, Configuring Handlers and Virtual Directories - Summary and self-assessment questions - SELF STUDY TOPIC: Advanced configurations for Azure Web Apps.

Module -2	Advanced Azure Configuration
Chapter 5	Azure SQL Database - Introduction/Overview of SQL Database - Comparing SQL Azure Database to IAAS / On-Premise SQL Server - Creating and Using SQL Server and SQL Database Services - Azure SQL Database Tools - Migrating on-premise database to SQL Azure - Elastic Pools, Backup and Recovery options in SQL Database - Summary and self-assessment questions - SELF STUDY TOPIC: Database migration strategies to Azure SQL.
Chapter 6	Microsoft Azure Storage - Introduction to storage - Storage types: file, disk, data lake storage, archive, cache, azure blob - Summary and self-assessment questions - SELF STUDY TOPIC: Implementing and managing Azure storage solutions.
Chapter 7	What is Microsoft Azure Used For - Microsoft azure used for App development, App hosting, software testing - Virtual machine creation, virtual hard drives - Used for AI and ML, DevOps - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring Azure for AI and ML deployments.
Chapter 8	Microsoft Azure Features - Improve backup and disaster Recovery - Develop and host web and mobile apps - Integration with active directory, Design for efficiency - Azure is designed for recoverability and scalable performance - Security - Summary and self-assessment questions - SELF STUDY TOPIC: Security best practices in Azure deployments.
Module -3	Specialized Azure Services
Chapter 9	Microsoft Azure Services - Azure application services: Azure AI, Azure Analytics, Azure IOT - Azure data services: Azure storage - Azure development services: Azure DevOps - Introduction to Azure computer services: Azure VMs, Azure Container - Azure network: Azure content delivery network (CDN) - Summary and self-assessment questions - SELF STUDY TOPIC: Integrating IoT with Azure for smart solutions.

Chapter 10	Azure Cloud Services - What is Cloud Service, Cloud Service vs App Service - Understand Cloud Service Roles - Cloud service models, Azure Backup, Using Azure Database in Cloud Service - Running Multiple Websites, Scaling a Cloud Service - Summary and self-assessment questions - SELF STUDY TOPIC: Scaling applications in Azure.
Chapter 11	Azure Active Directories - Introduction to azure active directory - Relationship between AD DS and Azure AD - Managing Users, Groups and Devices, Configuring Role Based Access Control - Implementing Azure AD B2B Collaboration - Implementing Azure AD B2C Collaboration - Summary and self-assessment questions - SELF STUDY TOPIC: Role-based access control in Azure.
Chapter 12	Managing PowerShell in Azure - Managing Azure Accounts and Subscriptions - Managing Resource Group, Managing App Service Plans and App Service Web Apps - Create and Configure a Storage Account, Managing Storage Accounts using PowerShell - Understanding Azure Resource Manager (ARM) - Summary and self-assessment questions - SELF STUDY TOPIC: Automation with Azure PowerShell.
Chapter 13	Microsoft Azure Cloud - IAAS - PAAS - SAAS - Summary and self-assessment questions - SELF STUDY TOPIC: Comparative study of IAAS, PAAS, and SAAS in Azure.
Chapter 14	Case Study - Azure Security Center case study - Azure data center case study - Summary and self-assessment questions - SELF STUDY TOPIC: Real-world applications and case studies on Azure.

d. Self-study topics for Advance learners:

- Deep dive into Azure architecture and its impact on cloud solutions.
- Implementing hybrid connectivity with Azure.
- Management and scalability of Azure VMs.
- Advanced configurations for Azure Web Apps.
- Database migration strategies to Azure SQL.
- Implementing and managing Azure storage solutions.

- Exploring Azure for AI and ML deployments.
- Security best practices in Azure deployments.
- Integrating IoT with Azure for smart solutions.
- Scaling applications in Azure.
- Role-based access control in Azure.
- Automation with Azure PowerShell.
- Comparative study of IAAS, PAAS, and SAAS in Azure.
- Real-world applications and case studies on Azure.

e. Textbooks / Reference Books

- Microsoft Azure Essentials Azure Web Apps for Developers by Rick Rainey, Azure Essentials.
- Azure for Architects: Implementing Cloud Design, DevOps, IoT, and Serverless Solutions on your Public Cloud by Ritesh Modi.
- Exam Ref AZ-900 Microsoft Azure Fundamentals by Jim Cheshire.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT704	Cloud Programming	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals, layers, and types of cloud computing.
2. To learn about cloud computing platforms, services, and enterprise architecture.
3. To explore various programming methods and models in cloud computing.
4. To gain hands-on experience in programming on cloud platforms like Google App Engine, Amazon AWS, and Microsoft Azure.

b. Course Outcomes

CO1	Understand the concepts, layers, and types of cloud computing.
CO2	Identify and compare different cloud computing platforms and services.
CO3	Apply various programming methods and models in cloud computing.
CO4	Develop applications using cloud services and SDKs on Google App Engine, Amazon AWS, and Microsoft Azure.
CO5	Deploy cloud services using HTTP and web services.

c. Syllabus

Module -1	Introduction to Cloud Computing
Chapter 1	Introduction to Cloud Computing - Overview - Layers and types of cloud - Cloud Computing Standards - Summary - Self-Assessment Questions - Self-Study Topics: Cloud Computing Architectures, Cloud Security
Chapter 2	Cloud Computing Platforms - Definition and Characteristics - Enterprise Computing - Internet Platforms - Cloud Computing Services - Enterprise Architecture - Summary - Self-Assessment Questions - Self-Study Topics: Hybrid Cloud, Multi-Cloud Environments
Chapter 3	Programming Methods in Cloud - Thread Programming - Task Reducing Programming - Map-Reduce Programming - Summary - Self-Assessment Questions - Self-Study Topics: Parallel Programming in Cloud, Distributed Programming in Cloud
Chapter 4	Cloud Programming with Software Environments - Features of Cloud with Grid Platforms - Parallel Programming Paradigms - Distributed Programming Paradigms - Summary - Self-Assessment Questions - Self-Study Topics: Cloud Programming Frameworks, Cloud IDE
Module -2	Cloud Programming Models and Platforms
Chapter 5	Cloud Programming Models - MapReduce - Spark - GraphLab and Samza - Spark Streaming - Summary - Self-Assessment Questions - Self-Study Topics: Flink, Storm

Chapter 6	MapReduce Programming Model - Execution Flow - Scheduling - Fault Tolerance - MapReduce Code using Apache Hadoop - Summary - Self-Assessment Questions - Self-Study Topics: YARN, Hadoop Ecosystem
Chapter 7	Introduction to Programming on Cloud - System Development Lifecycle - Working with Cloud SDK - Errors and Exception - Application and Infrastructure Monitoring - Self-Study Topics: Cloud Deployment Models, Cloud Migration Strategies
Chapter 8	Programming the Google App Engine - Google File System (GFS) - Bigtable - Google's NOSQL system - Chubby - Google's Distributed Lock Service - Summary - Self-Assessment Questions - Self-Study Topics: Google Cloud Datastore, Google Cloud Pub/Sub
Module-3	Programming on Cloud Platforms
Chapter 9	Programming Google App Engine with Python - Real Cloud Application on HTTP - Managing Data in the Cloud - Google App Engine Services for Login Authentication - Organizing Code by Separating UI and Logic - Summary - Self-Assessment Questions - Self-Study Topics: Google Cloud Endpoints, Google Cloud Functions
Chapter 10	Programming Google App Engine with Java - Google App Engine and Java - Managing Server-Side Data - Building User Interfaces in Java - Building the Server Side of a Java Application - Summary - Self-Assessment Questions - Self-Study Topics: Google Cloud Dataflow, Google Cloud Spanner

Chapter 11	Programming on Amazon (AWS) and Microsoft Azure - Programming on Amazon EC2 - Amazon Simple Storage Service S3 - Amazon Elastic Block Store EBS and Simple DB - Microsoft Azure Programming Support - Summary - Self-Assessment Questions - Self-Study Topics: AWS Lambda, Azure Functions
Chapter 12	Cloud Software Environments - Emerging Cloud Software Environments - Open-Source Eucalyptus and Nimbus - OpenNebula - Sector/Sphere - OpenStack - Manjrasoft Aneka Cloud - Summary - Self-Assessment Questions - Self-Study Topics: Docker, Kubernetes
Chapter 13	Applications using Cloud Services - Moving Applications to Cloud - Microsoft Cloud Services - Google Cloud Applications - Amazon Cloud Services - Cloud Applications - Summary - Self-Assessment Questions - Self-Study Topics: Serverless Computing, Cloud-Native Applications
Chapter 14	Deployment of Cloud Service - Deployment using HTTP - Deployment using Web Services - Self-Study Topics: Continuous Integration and Deployment in Cloud, Canary Deployments

d. Self-study topics for Advance learners:

Cloud Computing Architectures, Cloud Security, Hybrid Cloud, Multi-Cloud Environments, Parallel Programming in Cloud, Distributed Programming in Cloud, Cloud Programming Frameworks, Cloud IDE, Flink, Storm, YARN, Hadoop Ecosystem, Cloud Deployment Models, Cloud Migration Strategies, Google Cloud Datastore, Google Cloud Pub/Sub, Google Cloud Endpoints, Google Cloud Functions, Google Cloud Dataflow, Google Cloud Spanner, AWS Lambda, Azure Functions, Docker, Kubernetes, Serverless Computing, Cloud-Native Applications, Continuous Integration and Deployment in Cloud, Canary Deployments.

e. Textbooks / Reference Books

1. Dan Marinescu, "Cloud Computing: Theory and Practice", Morgan Kaufmann, 2017.
2. Thomas Erl, Robert Cope, Amin Naserpour, "Cloud Computing Design Patterns", Prentice Hall, 2015.
3. Dan C. Marinescu, "Cloud Computing: Theory and Practice", Elsevier, 2018.
4. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, "Mastering Cloud Computing: Foundations and Applications Programming", Morgan Kaufmann, 2013.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT705	Cloud Virtualization	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the fundamental concepts and architecture of cloud virtualization.
2. To explore different types of virtualization and their implementations in the cloud.
3. To analyze the challenges and security considerations in virtualized environments.

b. Course Outcomes

CO1	Understand the basic principles and architecture of cloud virtualization.
CO2	Differentiate between various types of virtualization techniques and their applications.
CO3	Identify and address common performance issues and security vulnerabilities in virtualized systems.
CO4	Apply virtualization technologies to optimize cloud infrastructure.
CO5	Evaluate and implement security measures for virtualization platforms.

c. Syllabus

Module -1	Fundamentals of Cloud Virtualization
Chapter 1:	Introduction to Virtualization - Introduction to cloud virtualization, Physical and virtual machines, Traditional and virtual computing - Understanding virtualization, Need and Applications of virtualization, Limitations - Challenges in Virtualized environment, tools and technologies in virtualized environments - Need of virtualization, cost - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring the evolution of virtualization technology and its impact on modern computing.
Chapter 2:	Types of Virtualization in Cloud - How virtualization works in cloud - Types of virtualization: Hardware virtualization (Full, emulation, and Para-virtualization), Implementation of hardware - Network virtualization, storage, server, data, application, desktop, OS virtualization - Summary and self-assessment questions - SELF STUDY TOPIC: In-depth study of para-virtualization and its benefits over full virtualization.
Chapter 3:	Architecture of Virtualization in Cloud - Architecture of cloud virtualization - Advantages and disadvantages of virtualization in cloud - Features of virtualization - Services in cloud virtualization - Summary and self-assessment questions - SELF STUDY TOPIC: Comparative analysis of virtualization architectures.
Chapter 4:	Server Virtualization - Introduction to server virtualization, server consolidation - Hypervisors, hypervisor types, architecture of hypervisors - Challenges in virtualization - Summary and self-assessment questions - SELF STUDY TOPIC: Case studies on hypervisor deployment in enterprise environments.
Module -2	Advanced Virtualization Concepts

Chapter 5:	Network and Memory Virtualization - Introduction to virtual LAN, IP addressing, memory addressing - Memory mapping, virtual memory, virtual memory configuration, VM scheduling - VM migrations and its types - Summary and self-assessment questions - SELF STUDY TOPIC: Techniques and challenges in virtual memory management.
Chapter 6:	Storage Virtualization - RAID, SCSI - Direct attached storage - Network attached storage - Storage area networks - Summary and self-assessment questions - SELF STUDY TOPIC: Best practices in managing storage virtualization.
Chapter 7:	Virtualization Issues - Virtualization performance issues - Hypervisor vulnerabilities, hypervisor attacks - VM attacks, VM migration attacks and security solutions - Summary and self-assessment questions - SELF STUDY TOPIC: Security strategies for mitigating virtualization-specific threats.
Chapter 8:	Virtualized Data Centers - Data center and its architecture - Role of virtualization in enabling the cloud - Cloud infrastructures; public, private, hybrid - Service provider interfaces; SaaS, PaaS, IaaS - Summary and self-assessment questions - SELF STUDY TOPIC: Evaluating the role of virtualization in cloud data center scalability and flexibility.
Module -3	Security and Case Studies in Virtualization
Chapter 9:	Virtualization System-Specific Attacks - Guest hopping, attacks on the VM (delete the VM, attack on the control of the VM, code or file injection into the virtualized file structure) - VM migration attack, hyper jacking. - Virtualization-based sandboxing - Summary and self-assessment questions - SELF STUDY TOPIC: Analysis of attack vectors in virtualized environments and mitigation techniques.

Chapter 10:	Virtualization Techniques - Hardware-assisted virtualization - Virtualization in VMX mode - I/O virtualization - Memory virtualization techniques, Para virtualization in Xen - Summary and self-assessment questions - SELF STUDY TOPIC: Exploring advanced virtualization techniques and their impact on performance and security.
Chapter 11:	Network Virtualization - VM checkpoint and cloning - Virtual local area network - Virtual extensible local area network, and generic routing encapsulation - Summary and self-assessment questions - SELF STUDY TOPIC: Deploying network virtualization in large scale environments.
Chapter 12:	Virtualization at Different Levels - Virtualization at OS level - Middleware support for Virtualization - Virtualization Tools and mechanisms - Virtualization in multi-core processors - Summary and self-assessment questions - SELF STUDY TOPIC: Middleware and OS-level virtualization techniques.
Chapter 13:	Security for Virtualization Platforms - Host security for SaaS, PaaS, IaaS - Data security and data confidentiality, data integrity, data encryption for cloud virtualization - Summary and self-assessment questions - SELF STUDY TOPIC: Comprehensive security practices for virtualized platforms.
Chapter 14:	Case Study - Private cloud implementation using Hyper-V and VMware - Public cloud implementation or Machine learning Virtualization case study - Summary and self-assessment questions - SELF STUDY TOPIC: Real-world implementations of virtualization technologies in different cloud settings.

d. Self-study topics for Advance learners:

- Evolution of virtualization technology and its impact on modern computing.
- In-depth study of para-virtualization and its benefits over full virtualization.
- Comparative analysis of virtualization architectures.
- Case studies on hypervisor deployment in enterprise environments.
- Techniques and challenges in virtual memory management.

- Best practices in managing storage virtualization.
- Security strategies for mitigating virtualization-specific threats.
- Evaluating the role of virtualization in cloud data center scalability and flexibility.
- Analysis of attack vectors in virtualized environments and mitigation techniques.
- Exploring advanced virtualization techniques and their impact on performance and security.
- Deploying network virtualization in large scale environments.
- Middleware and OS-level virtualization techniques.
- Comprehensive security practices for virtualized platforms.
- Real-world implementations of virtualization technologies in different cloud settings.

e. Textbooks / Reference Books

1. Virtualization Essentials by Matthew Portnoy.
2. Mastering Virtualization by Tim Warner.
3. Cloud Computing: Concepts, Technology & Architecture by Thomas Erl.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus: Semester III - Full Stack Development

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT706	HTML, CSS and Javascript	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the fundamental concepts and technologies used in web development.
2. To develop skills in HTML, CSS, and JavaScript for building and styling web pages.
3. To explore advanced web development techniques and tools for creating interactive and dynamic websites.

b. Course Outcomes

CO1	Understand the structure and functioning of the internet and web development.
CO2	Utilize HTML to create structured web pages.
CO3	Apply CSS for styling and layout adjustments.
CO4	Implement JavaScript to add interactivity to web pages.
CO5	Demonstrate the ability to develop a complete website from planning to deployment.

c. Syllabus

Module -1	Foundations of Web Development
Chapter 1	Introduction to Web Development & Related Terms - Overview of Web, Internet - Requirement of Internet - Application areas of Internet - Key difference between Web & Internet - Client - server computing - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The evolution of web technologies and their impact on modern computing.
Chapter 2	Introduction to Website Development - Meaning of web page - Website - Basic types of websites - Types of website development - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Comparative analysis of static and dynamic websites.
Chapter 3	Hypertext Markup Language -I - Introduction to Hypertext Markup Language - Uses of HTML - Limitations of HTML - Concept of Tag, Attributes - Denotation of tags - Basic structure of HTML Program - Text formatting tags - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The role of HTML in website accessibility.
Chapter 4	Hypertext Markup Language -II - List Tag & its attributes - Adding Image in a Webpage with its all attributes - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Enhancing user experience through advanced HTML tagging.
Module -2	Advanced Markup and Styles

Chapter 5	HyperText Markup Language -III - ImageMap - Table Tag & its attributes - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Designing responsive tables and image maps for various devices.
Chapter 6	HyperText Markup Language -IV - Linking of document using Anchor Tag - Types of Linking - Controls of form - GET & POST Method - Dividing a screen into multiple sections/frames using Frameset Tag - Targeted frame - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Effective form design and management in modern web applications.
Chapter 7	HTML -5 - Key features of HTML-5 - Tags Graphics in HTML5 - Media tags in HTML5 - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Leveraging HTML5 for multimedia integration.
Chapter 8	Cascading Style Sheet (CSS) - Introduction to CSS - Use of CSS - Types of CSS - Selectors, Properties, Values - CSS Properties - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advanced CSS techniques for cross-browser compatibility.
Module -3	JavaScript and Dynamic Interactions

Chapter 9	JavaScript -I - Introduction to JavaScript - JavaScript Variables, Data types, Operators - Built-in functions in JavaScript - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Exploring the execution context and scope in JavaScript.
Chapter 10	JavaScript -II - Control structure in JavaScript - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Debugging and error handling in JavaScript.
Chapter 11	JavaScript -III - DOM, Math, Array, History, Navigator, Location, Windows, String, Date, Document objects - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Manipulating page content and structure using DOM.
Chapter 12	JavaScript -IV - Functions in JavaScript, types of functions - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Implementing modular JavaScript code.
Chapter 13	JavaScript -V - Events in JavaScript, event handling in JavaScript - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advanced event handling techniques in complex web applications.
Chapter 14	Steps to Create a Website - Requirement for any website creation - Steps to create a website - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Project management strategies for web development projects.

d. Self-study topics for Advance learners:

- Evolution of web technologies and their impact on modern computing.
- Comparative analysis of static and dynamic websites.
- The role of HTML in website accessibility.
- Enhancing user experience through advanced HTML tagging.

- Designing responsive tables and image maps for various devices.
- Effective form design and management in modern web applications.
- Leveraging HTML5 for multimedia integration.
- Advanced CSS techniques for cross-browser compatibility.
- Exploring the execution context and scope in JavaScript.
- Debugging and error handling in JavaScript.
- Manipulating page content and structure using DOM.
- Implementing modular JavaScript code.
- Advanced event handling techniques in complex web applications.
- Project management strategies for web development projects.

e. Textbooks / Reference Books

1. HTML & CSS: Design and Build Websites by Jon Duckett.
2. JavaScript and JQuery: Interactive Front-End Web Development by Jon Duckett.
3. Eloquent JavaScript: A Modern Introduction to Programming by Marijn Haverbeke.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT707	User Interface, Experience, Design	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the fundamental concepts of user interface design and human-computer interaction.
2. To explore various user interface types including graphical and web interfaces.
3. To develop skills in designing effective user interfaces with a focus on usability and accessibility.

b. Course Outcomes

CO1	Understand the basic principles of user interface design and human-computer interaction.
CO2	Design graphical user interfaces that are both functional and aesthetically pleasing.
CO3	Apply principles of web interface design effectively in creating user-friendly web pages.
CO4	Evaluate and utilize different tools and principles for improving interface design.
CO5	Incorporate accessibility and internationalization considerations in design to cater to diverse user groups.

c. Syllabus

Module -1	Basics of User Interface Design
Chapter 1:	Introduction to User Interface - Introduction - User Interface - Important Elements of a User Interface - Software for user interface - Good User Interface - Human Computer Interface (HCI) - Roots of HCI in India - History of Screen Design - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Historical development of HCI and its impact on modern interfaces.
Chapter 2:	The Graphical User Interface - Introduction - Graphical User Interface - Working of Graphical User Interface - The Evolution of Graphic Design - Direct manipulation - Graphical system advantages & Disadvantages - Characteristics of the Graphical User Interface - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Analyzing the evolution of graphical user interfaces from Xerox PARC to modern frameworks.
Chapter 3:	The Web User Interface - Introduction - The Web User Interface - Characteristics of a Web Interface - Difference between GUI and Web Interface Design - Merging Graphical business system and Web - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Comparisons of user interaction patterns in GUI and web interfaces.

Chapter 4:	Principles of User Interface Design - Introduction - Principles of User Interface Design - Principles for the Xerox STAR - General principles of User-Interface - Designing for Performance Quality - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Application of classic UI design principles in contemporary software development.
Chapter 5:	Know Your User or Client and Understand the Business Function - Introduction - Understanding How People Interact with Computers - Responses to Poor Design - Important Human Characteristics in Design - Human Considerations in the Design of Business Systems - Human Interaction Speeds - Methods for Gaining an Understanding of Users - Business Definition and Requirements Analysis - Determining Basic Business Functions - System Training and Documentation Needs - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Methods and tools for effective user research in UI design.
Module -2	Advanced User Interface Design Concepts
Chapter 6:	Understand the Principles of Good Interface and Screen Design - Introduction - Human Considerations in Interface and Screen Design - Interface Design Goals - Technological Considerations in Interface Design - Examples of Screens - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Integrating human factors into digital interface design.

Chapter 7:	Develop System Menus and Navigation Schemes - Introduction - Structures of Menus - Functions of Menus - Formatting of Menus - Phrasing the Menu - Kinds of Graphical Menus - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Designing intuitive navigation schemes for enhanced user experiences.
Chapter 8:	Select the Proper Kinds of Windows - Introduction - Window Characteristics - Components of a Window - Window Presentation Styles - Types of Windows - Organizing Window Functions - Window Organization - General Operations on Windows - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Exploring the role of window management in multitasking environments.
Chapter 9:	Choose the Proper Kinds of Device and Screen based controls - Introduction - Characteristics of Device-Based Controls - Selecting the Proper Input Device - Guidelines for Selecting the Proper Input Device - Operable Controls - Selection Method and Indication - Other Operable Controls - Presentation Controls - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Evaluating the effectiveness of different input devices in various use cases.

Chapter 10:	Write Clear Text and Messages & Provide Effective Feedback and Guidance and Assistance - Introduction - Words, Sentences, Messages, and Text - Writing Message Box Text - Words - Response Time - Guidance and Assistance - Problem Management - Providing Guidance and Assistance - Problems with Documentation - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Best practices in user communication and documentation within interfaces.
Module -3	Specialized Topics in User Interface Design
Chapter 11:	Provide Effective Internationalization and Accessibility - Introduction - International Considerations - Localization - Cultural Considerations - Words and Text - Images and Symbol - Accessibility - Types of Disabilities - Accessibility Design - Visual Disabilities - Hearing Disabilities - Cursor - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Strategies for creating universally accessible and culturally sensitive interfaces.

Chapter 12:	Create Meaningful Graphics, Icons and Images & Choosing the Proper Colors - Introduction - Icons - Characteristics of Icons - Influences on Icon Usability - Size - Choosing Icon Images - Creating Icon Images - Multimedia - Graphics - Color – What Is It? - Possible Problems with Color - Color and Human Vision - Choosing Colors - Colors in Context - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The impact of color psychology on user perception and interface usability.
Chapter 13:	Organize and Layout Windows and Pages - Introduction - Organizing and Laying Out Screens - Organization Guidelines - Screen Examples - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advanced layout techniques for optimizing user interface flow and aesthetics.
Chapter 14:	Test, Test, and Retest - Introduction - The Purpose of Usability Testing - Scope of Testing - Prototype - Hand Sketches and Scenarios - Kinds of Tests - Developing and Conducting a Test - Analyze, Modify, and Retest - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Developing a comprehensive testing regimen for iterative design improvements.

d. Self-study topics for Advance learners:

- Historical development of HCI and its impact on modern interfaces.
- Analyzing the evolution of graphical user interfaces from Xerox PARC to modern frameworks.
- Comparisons of user interaction patterns in GUI and web interfaces.
- Application of classic UI design principles in contemporary software development.
- Methods and tools for effective user research in UI design.
- Integrating human factors into digital interface design.
- Designing intuitive navigation schemes for enhanced user experiences.
- Exploring the role of window management in multitasking environments.
- Evaluating the effectiveness of different input devices in various use cases.
- Best practices in user communication and documentation within interfaces.
- Strategies for creating universally accessible and culturally sensitive interfaces.
- The impact of color psychology on user perception and interface usability.
- Advanced layout techniques for optimizing user interface flow and aesthetics.
- Developing a comprehensive testing regimen for iterative design improvements.

e. Textbooks / Reference Books

1. The Design of Everyday Things by Don Norman.
2. Don't Make Me Think by Steve Krug.
3. About Face: The Essentials of Interaction Design by Alan Cooper.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT708	DevOps -1 (GIT, Jenkins, Docker)	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of DevOps, its history, goals, and stakeholders.
2. To explore various SDLC models, software testing methodologies, and Agile practices.
3. To gain hands-on experience with version control using Git and continuous integration using Jenkins.
4. To learn about Docker, its architecture, and installation for containerization.

b. Course Outcomes

CO1	Understand the concepts, principles, and ecosystem of DevOps.
CO2	Apply Agile methodologies and software testing techniques in the DevOps lifecycle.
CO3	Utilize Git for version control and collaborate with remote repositories.
CO4	Implement continuous integration and deployment using Jenkins.
CO5	Understand the fundamentals of Docker and its use cases in containerization.

c. Syllabus

Module -1	Introduction to DevOps and Agile Methodologies
Chapter 1	Introduction to DevOps - Introduction to DevOps - What is DevOps? - SDLC models, Lean, ITIL, Agile - Why DevOps? - History of DevOps - DevOps Stakeholders - DevOps Goals - Important terminology - DevOps perspective - DevOps and Agile - DevOps Tools - Configuration management - Continuous Integration and Deployment - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 2	Overview of DevOps - Why DevOps? - DevOps Market Trends - DevOps Engineer Skills - DevOps Delivery Pipeline - DevOps Ecosystem - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 3	Introduction to SDLC, Software testing, Agile : Software testing lifecycle - Working with Block box testing - Working with White box testing - Working Grey box testing - Working with Function testing - Working with Regressing testing, smoke testing, System testing, Integration testing etc. - Summary - Self-Assessment Questions - References/Reference Reading

Chapter 4	Agile Methodologies - Process flow of Scrum Methodologies - Project planning, scrum testing, sprint Planning and Release management - Analysis - Design, Execution and wrapping closure - Summary - Self-Assessment Questions - References/Reference Reading
Module -2	Version Control with Git and Continuous Integration
Chapter 5	Introduction to Git - Introduction of Git - Selecting Git Client - Creating Repository - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 6	Version Control with Git - What is version control - What is Git - Why Git for your organization - Install Git - Common commands in Git - Working with Remote Repositories - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 7	Overview of GIT - Overview of SVN, GIT, Clear case, perforce & Comparision - Working with Tag - Creating and Merging Branches - Executing Git Commands - Git Logs, Git stash, Git rebase - Merge conflict issues resolving - Git pull, clone, fetch - Summary - Self-Assessment Questions - References/Reference Reading

Chapter 8	Git Integration - Branching and Merging in Git - Git workflows - Git cheat sheet - What is CI - Why CI is Required - Summary - Self-Assessment Questions - References/Reference Reading
Module -3	Jenkins and Docker
Chapter 9	Introduction to Jenkins - Introduction to Jenkins (With Architecture) - What is Continuous Integration - Jenkins Continuous Integration - What is Continuous Deployment - Jenkins Vs Jenkins Enterprise - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 10	Jenkins Installation - Downloading and Installing Jenkins using TomCat - Creating Jenkins as a Service. - Starting and Stopping Jenkins - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 11	Configure Jenkins and User Management - Secure Jenkins - Create a new user - Generate ssh key for Jenkins user - Plug-in management - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 12	Introduction to Docker - What is a Docker - Use case of Docker - Platforms for Docker - Dockers vs. Virtualization - Summary - Self-Assessment Questions - References/Reference Reading

Chapter 13	Architecture of Docker - Docker Architecture - Understanding the Docker components - Summary - Self-Assessment Questions - References/Reference Reading
Chapter 14	Docker Installation - Creating a Virtual Docker Host(CentOS) by using Vagrant - Installing Docker on CentOS - Introduction to Docker namespaces - Summary - Self-Assessment Questions - References/Reference Reading
Chapter	

d. Self-study topics for Advance learners:

DevOps Culture, DevOps Maturity Model, DevOps and Cloud Computing, DevOps and Microservices, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Kanban, Lean Software Development, Git Submodules, Git Hooks, Jenkins Pipelines, Jenkins Shared Libraries, Docker Compose, Docker Swarm, Kubernetes.

e. Textbooks / Reference Books

1. Gene Kim, Jez Humble, Patrick Debois, John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations", IT Revolution Press, 2016.
2. Jon Loeliger, Matthew McCullough, "Version Control with Git: Powerful tools and techniques for collaborative software development", O'Reilly Media, 2012.
3. Brent Laster, "Jenkins 2: Up and Running: Evolve Your Deployment Pipeline for Next Generation Automation", O'Reilly Media, 2018.
4. Sean P. Kane, Karl Matthias, "Docker: Up & Running: Shipping Reliable Containers in Production", O'Reilly Media, 2018.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT709	Software Architecture	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce fundamental concepts and importance of software architecture in system design.
2. To explore various architectural patterns, styles, and quality attributes that influence software design.
3. To examine case studies that demonstrate the application of architectural principles in real-world systems.

b. Course Outcomes

CO1	Understand and describe different software architectural patterns and styles.
CO2	Analyze software architecture's role in ensuring system functionality and quality.
CO3	Apply architectural patterns to solve specific design problems in software development.
CO4	Evaluate architectural decisions based on quality attributes and system requirements.
CO5	Utilize formal models and specifications for defining and documenting software architecture.

c. Syllabus

Module -1	Core Concepts of Software Architecture
Chapter 1:	Introduction To Software Architecture - Introduction - Architectural Patterns - Reference Models and Reference - Importance of Software Architecture - Architectural Structures and Views - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Evolution of software architecture as a discipline.
Chapter 2:	Architectural Styles - Architectural Styles - Other Familiar Architectures - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Comparative analysis of different architectural styles and their applicability.
Chapter 3:	Software Architecture – Case Studies - Key Word in Context - Instrumentation Software - Mobile Robotics, Cruise Control - Three Vignettes in Mixed Style - Real Time Applications and Distributed Applications - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Impact of architectural decisions in the provided case studies.
Chapter 4:	Architectural Quality Attributes - Functionality and Architecture - Architecture and Quality Attributes - System Quality Attributes - Quality Attributes Scenario in Practice - Other System Quality Attributes - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Strategies for balancing competing quality attributes in software design.

Chapter 5:	Achieving Quality - Introduction - Tactics - Relationship of Tactics to Architectural Patterns - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Methods for achieving system quality through architectural tactics.
Module -2	Advanced Architectural Concepts
Chapter 6:	Architectural Patterns – 1 - Architectural Pattern - From Mud to Structure - Layers - Pipes and filters - Blackboard - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Application of layering and filters in software architecture.
Chapter 7:	Architectural Patterns – 2 - Distributed Systems - Broker architecture - Interactive Systems - Model-View-Controller (MVC) - Presentation-Abstraction-Control (PAC) - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The role of MVC in modern web application development.
Chapter 8:	Architectural Patterns – 3 - Adaptable Systems - Microkernel - Reflection - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Benefits and challenges of using Microkernel in system design.
Chapter 9:	Important Design Patterns - Design Patterns - Structural Decomposition - Organization of Work - Access Control - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Exploring design patterns and their impact on software modularity and maintenance.

Chapter 10:	Architectural Design Guidance - User Interface Architecture - The Quantified Design Space - Architectural Design Space Formalism - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Formal methods in architectural design to quantify design decisions.
Module -3	Specialized Architectural Knowledge
Chapter 11:	Formal Models and Specifications - Z-Notation - Formalizing an Architectural Style - Formalizing an Architectural Design Space - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Utilization of formal specifications in defining architectural styles.
Chapter 12:	Linguistic Issues - Architectural Description Language - First Class Connectors - Adding Implicit Invocation to Traditional Programming Languages - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Enhancing architectural languages to support system evolution.
Chapter 13:	Tools for Software Architecture - CASE Tools - Analysis and Design tools - Software Development Tools - Software Tools for Architecture Design - Excel as an Architecture Tool - Exploiting Style in Architectural Design - Quality-Driven Software Architecture Design - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Evaluation of different software tools for architecture design and their effectiveness.

Chapter 14:	Designing and Documenting Software Architecture - Forming a Team Structure - Creating a Skeleton System - Uses of Architectural Documentation - Rules for Documentation - Views - Documenting a View - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Best practices in architectural documentation for clarity and maintenance.
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d. Self-study topics for Advance learners:

- Evolution of software architecture as a discipline.
- Comparative analysis of different architectural styles and their applicability.
- Impact of architectural decisions in the provided case studies.
- Strategies for balancing competing quality attributes in software design.
- Methods for achieving system quality through architectural tactics.
- Application of layering and filters in software architecture.
- The role of MVC in modern web application development.
- Benefits and challenges of using Microkernel in system design.
- Exploring design patterns and their impact on software modularity and maintenance.
- Formal methods in architectural design to quantify design decisions.
- Utilization of formal specifications in defining architectural styles.
- Enhancing architectural languages to support system evolution.
- Evaluation of different software tools for architecture design and their effectiveness.
- Best practices in architectural documentation for clarity and maintenance.

e. Textbooks / Reference Books

1. Software Architecture in Practice by Len Bass, Paul Clements, and Rick Kazman.
2. Patterns of Enterprise Application Architecture by Martin Fowler.
3. Documenting Software Architectures: Views and Beyond by Paul Clements et al.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT710	Prototyping	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the concept of design thinking and how it applies to prototyping.
2. To explore various types of prototyping techniques and their appropriate applications.
3. To develop skills in creating, evaluating, and refining prototypes for different stages of product development.

b. Course Outcomes

CO1	Define prototyping and discuss its role in the design and development process.
CO2	Identify different prototyping techniques and determine when each should be used.
CO3	Develop and evaluate prototypes using a range of methods to ensure usability and effectiveness.
CO4	Apply best practices in prototyping to enhance communication and collaboration among stakeholders.
CO5	Understand the tools and technologies available for creating interactive and high-fidelity prototypes.

c. Syllabus

Module -1	Fundamentals of Prototyping
Chapter 1:	Introduction to Prototyping - What is Design Thinking? - What is Prototyping? - Why use Prototypes? - Different kinds of Prototype - How to create a Prototype - Advantages of Prototyping - Uses of Prototyping - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The role of prototyping in accelerating innovation in product development.
Chapter 2:	Prototyping Types - Rapid (Throwaway) prototyping. - Evolutionary prototyping. - Incremental prototyping. - Extreme prototyping. - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advantages and limitations of different prototyping models.
Chapter 3:	Prototype Evaluation - Protocol Analysis - Communicability Evolution. - User Participation. - Rapid Ethnography. - Experience Prototyping - Tools - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Effective methods for user testing and feedback collection.
Chapter 4:	Practical Look at Prototyping - Five reasons why you need to prototype. - How prototypes improve collaboration and communication. - How prototypes add balance to design. - How prototyping makes usability testing easier. - Why prototyping is mandatory for mobile - Takeaway - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Impact of prototyping on team dynamics and project success.

Chapter 5:	Choosing the Right Prototyping Process & Fidelity - When to Start Prototyping: 3 Points of Convergence - How to Prototype: The Rapid Prototyping Process - What Is a Prototype: The 4 Dimensions? - What to Prototype: 4 Ways to Combine Fidelity & Functionality. - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Balancing fidelity and functionality in prototype design.
Module -2	Advanced Prototyping Techniques
Chapter 6:	Traditional Prototyping Methods and Tools - Paper Prototypes - Wizard of Oz Prototypes - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Exploring low-fidelity prototyping and its effectiveness in early design stages.
Chapter 7:	Digital Prototyping Methods and Tools - Presentation Software - Coded (HTML) Prototype - Prototyping Software & Apps - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Comparison of digital prototyping tools and their use cases.
Chapter 8:	Creating Prototypes for Usability Testing - Knowing Your Users: Personas, Scenarios, and Experience Maps - Usability Tests before the Prototype - The Right Users and the Right Tasks - General Advice for Testing Prototype Usability - Different Fidelities for Testing Prototypes - Content Guidelines for Testing Any Prototype - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Strategies for integrating usability testing into the prototyping process.

Chapter 9:	Prototyping Best Practices - Know Your Audience and Your Goals - Prime Your Audience Beforehand - Involve the Users (Participatory Design) - Focus on Flows and User Scenarios - Keep Clicking Simple - Don't Neglect Animations - Sketching: The Prototype for the Prototype - Don't Let Coding Hold You Back - Use Prototypes for Usability Tests - Prototype Only What You Need – Then Stop - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Effective use of animations and interactions in prototypes.
Chapter 10:	Wireframing & Prototyping: Past, Present, and Future - Present: The Current State of Design - Present: The Current State of Prototyping - Past: A Prototyping Timeline - Future: The Age of Prototyping - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Evolution of prototyping tools and techniques over the decades.
Module -3	Prototyping in Modern Application Development
Chapter 11:	4th Generation Systems - Object-Oriented Application Frameworks - Web Applications Prototyping - Benefits - Final Thoughts - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The impact of object-oriented frameworks on rapid prototyping.
Chapter 12:	Creating Interactive Prototypes from Photoshop Files-I - Importing from Photoshop – overview video - Importing from Photoshop to step by step - General notes on using interactions and animations - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Techniques for creating highly interactive prototypes from static design files.

Chapter 13:	Creating Interactive Prototypes from Photoshop Files-II - Button: scrolling the page after click, changing style on hover - Form: triggering visibility on scroll - Form: interactive inputs - Form: interactions after signing up - Previewing and gathering feedback - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advanced interaction techniques for enhancing user engagement.
Chapter 14:	How to Create Interactive Prototypes from Sketch Files - Overview Video - Importing from Sketch - Prototyping Animations & Interactions - Previewing Animations & Interactions - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Leveraging Sketch for rapid prototyping and iterative design.

d. Self-study topics for Advance learners:

- The role of prototyping in accelerating innovation in product development.
- Advantages and limitations of different prototyping models.
- Effective methods for user testing and feedback collection.
- Impact of prototyping on team dynamics and project success.
- Balancing fidelity and functionality in prototype design.
- Exploring low-fidelity prototyping and its effectiveness in early design stages.
- Comparison of digital prototyping tools and their use cases.
- Strategies for integrating usability testing into the prototyping process.
- Effective use of animations and interactions in prototypes.
- Evolution of prototyping tools and techniques over the decades.
- The impact of object-oriented frameworks on rapid prototyping.
- Techniques for creating highly interactive prototypes from static design files.
- Advanced interaction techniques for enhancing user engagement.
- Leveraging Sketch for rapid prototyping and iterative design.

e. Textbooks / Reference Books

1. Prototyping and Modelmaking for Product Design by Bjarki Hallgrimsson.
2. The Art of Innovation: Lessons in Creativity from IDEO, America's Leading Design Firm by Tom Kelley.

3. Don't Make Me Think: A Common Sense Approach to Web Usability by Steve Krug.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus: Semester III - Data Analytics

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT711	Data Analytics Using Python	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the fundamentals of data analytics and its applications using Python.
2. To explore statistical techniques for data analysis including probability distributions, hypothesis testing, and regression analysis.
3. To develop practical skills in data manipulation, visualization, and analysis using Python libraries such as Pandas, Matplotlib, and Scikit-learn.

b. Course Outcomes

CO1	Understand the role of data analytics in decision-making and problem-solving.
CO2	Apply statistical methods to analyze and interpret data using Python.
CO3	Utilize Python for creating descriptive and inferential statistical models.
CO4	Conduct regression analysis, ANOVA, and hypothesis testing to draw conclusions from data.
CO5	Implement machine learning algorithms for clustering and classification.

c. Syllabus

Module -1	Statistical Foundations and Python Programming
Chapter 1	Introduction to Data Analytics and Python Fundamentals - Introduction to Data Analytics - Python Fundamentals - Central Tendency and Dispersion - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The evolution of data analytics in the information age.
Chapter 2	Introduction to Probability - Introduction to Probability - Probability Distributions - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Real-world applications of probability distributions in data analysis.
Chapter 3	Sampling and Sampling Distributions - Python Demo for Distributions - Sampling and Sampling Distribution - Distribution of Sample Means, population, and variance - Confidence interval estimation: Single population - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The importance of sampling methods in data science.
Chapter 4	Hypothesis Testing - Hypothesis Testing - Errors in Hypothesis Testing - Hypothesis Testing: Two sample test - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Impact of hypothesis testing in medical research and quality control.
Module -2	Advanced Statistical Techniques
Chapter 5	Introduction to ANOVA - ANOVA - Two Way ANOVA - Post Hoc Analysis (Tukey's Test) - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Comparing means from more than two groups in agricultural experiments.

Chapter 6	Linear Regression - Randomize block design (RBD) - Linear Regression - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Applications of linear regression in economic forecasting.
Chapter 7	Multiple Regression - Estimation, Prediction of Regression Model Residual Analysis - Multiple Regression Model - Categorical variable regression - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Use of multiple regression in marketing analytics and customer behavior modeling.
Chapter 8	Concepts of MLE and Logistic Regression - Maximum Likelihood Estimation - Logistic Regression - Linear Regression Model Vs Logistic Regression Model - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Advantages of logistic regression over linear regression in binary outcome modeling.
Module -3	Data Visualization and Machine Learning
Chapter 9	ROC and Regression Analysis Model Building - Confusion matrix and ROC - Performance of Logistic Model - Regression Analysis Model Building - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Evaluating model performance with ROC curves in clinical diagnostics.
Chapter 10	Chi-Square Test - Chi - Square Test of Independence - Chi-Square Goodness of Fit Test - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Application of chi-square tests in genetics and marketing research.

Chapter 11	Data Visualization with Python - Introduction to Matplotlib - Introduction to Seaborn - Plotting data in Python using Matplotlib and Seaborn - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The impact of visual elements on the interpretation of data insights.
Chapter 12	Data Manipulation using Pandas Techniques - Introduction - Different Pandas techniques - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Streamlining data cleaning processes with Pandas for large datasets.
Chapter 13	Clustering Analysis - Clustering analysis - K- Means Clustering - Hierarchical method of clustering - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Applications of clustering in market segmentation and image segmentation.
Chapter 14	Classification and Regression Trees (CART) - Introduction to Scikit Learn - Different types of Classifiers - Classification and Regression Trees - Measures of attribute selection - Attribute selection Measures in CART - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The role of decision trees in financial risk assessment and customer retention strategies.

d. Self-study topics for Advance learners:

- The evolution of data analytics in the information age.
- Real-world applications of probability distributions in data analysis.
- The importance of sampling methods in data science.
- Impact of hypothesis testing in medical research and quality control.
- Comparing means from more than two groups in agricultural experiments.
- Applications of linear regression in economic forecasting.
- Use of multiple regression in marketing analytics and customer behavior modeling.
- Advantages of logistic regression over linear regression in binary outcome modeling.
- Evaluating model performance with ROC curves in clinical diagnostics.

- Application of chi-square tests in genetics and marketing research.
- The impact of visual elements on the interpretation of data insights.
- Streamlining data cleaning processes with Pandas for large datasets.
- Applications of clustering in market segmentation and image segmentation.
- The role of decision trees in financial risk assessment and customer retention strategies.

e. Textbooks / Reference Books

1. Python for Data Analysis by Wes McKinney.
2. Data Science from Scratch: First Principles with Python by Joel Grus.
3. Practical Statistics for Data Scientists by Peter Bruce and Andrew Bruce.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT712	SQL for Data Analytics	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of SQL and its role in managing relational databases.
2. To learn and apply SQL commands for creating, modifying, and querying tables.
3. To utilize subqueries, aggregate functions, and joins for advanced data analysis.
4. To explore data cleaning techniques, window functions, and complex data types for analytics using SQL.
5. To optimize SQL performance through query planning, index scanning, and other techniques.

b. Course Outcomes

CO1	Understand the basics of relational databases, SQL, and various types of SQL statements.
CO2	Create, modify, and manage tables using SQL commands and constraints.
CO3	Apply subqueries, aggregate functions, and joins for advanced data analysis.
CO4	Perform data cleaning, utilize window functions, and handle complex data types for analytics using SQL.
CO5	Optimize SQL performance through query planning, index scanning, and other techniques.

c. Syllabus

Module -1	Introduction to SQL and Table Management
Chapter 1	Introduction to SQL - Introduction - Definition of DATA - Data Types - Relational Database and SQL - Types of SQL Statement - DDL, DML, DQL, DCL, CONSTRAINTS - Summary - Self-Assessment Questions - Self-Study Topics: SQL History, SQL vs. NoSQL
Chapter 2	Managing Tables - Creating tables - Creating tables using select statement - Adding and removing columns - Removing table - Adding, updating and deleting data - Summary - Self-Assessment Questions - Self-Study Topics: Temporary Tables, Table Partitioning
Chapter 3	SQL Basics for Analytics - Select statement - Where clause - Using And, Or, In, Not In, Like, Wildcards - Summary - Self-Assessment Questions - Self-Study Topics: Null Values, Distinct Keyword
Chapter 4	Subqueries - Introduction - Single Row Subquery - Multiple Row Subquery - Summary - Self-Assessment Questions - Self-Study Topics: Correlated Subqueries, EXISTS Operator
Module -2	Advanced SQL Concepts and Functions
Chapter 5	Aggregate Functions - SUM, COUNT, AVG, MIN, MAX, VARIANCE - GROUP BY - HAVING CLAUSE - Summary - Self-Assessment Questions - Self-Study Topics: ROLLUP, CUBE

Chapter 6	Functions - STRING - NUMERIC - DATE - CASE - Summary - Self-Assessment Questions - Self-Study Topics: User-Defined Functions, Regular Expressions
Chapter 7	Joins - Introduction - Types of Joins (EQUI, INNER, OUTER, LEFT, RIGHT) - UNIONS - Summary - Self-Assessment Questions - Self-Study Topics: Self Join, Cross Join
Chapter 8	Views and Indexing - Concept - Creating View from Single, Multiple Tables - INDEX: Concept and Advantages - Summary - Self-Assessment Questions - Self-Study Topics: Materialized Views, Clustered vs. Non-Clustered Indexes
Module -3	Data Cleaning, Analytics, and Performance Optimization
Chapter 9	Data Cleaning - Introduction - Clean Up with SQL - Summary - Self-Assessment Questions - Self-Study Topics: Data Normalization, Data Validation
Chapter 10	Windows Functions - Introduction - WINDOW Keyword - WINDOW Functions - Summary - Self-Assessment Questions - Self-Study Topics: RANK, DENSE_RANK, LEAD, LAG
Chapter 11	Analytics Using Complex Data Types - Introduction - DATE and TIME Datatypes for Analysis - Summary - Self-Assessment Questions - Self-Study Topics: JSON Data Type, XML Data Type

Chapter 12	Importing Exporting Data - Introduction - The COPY Command - Summary - Self-Assessment Questions - Self-Study Topics: SQL Loader, Data Pump
Chapter 13	Functions and Triggers - Functions with and without Argument - Creating Triggers - Summary - Self-Assessment Questions - Self-Study Topics: Stored Procedures, Event Scheduling
Chapter 14	Performant SQL - Database Scanning Method - Query Planning - Index Scanning - Summary - Self-Assessment Questions - Self-Study Topics: Query Optimization Techniques, SQL Profiling

d. Self-study topics for Advance learners:

- SQL History, SQL vs. NoSQL, Temporary Tables, Table Partitioning, Null Values, Distinct Keyword, Correlated Subqueries, EXISTS Operator, ROLLUP, CUBE, User-Defined Functions, Regular Expressions, Self Join, Cross Join, Materialized Views, Clustered vs. Non-Clustered Indexes, Data Normalization, Data Validation, RANK, DENSE_RANK, LEAD, LAG, JSON Data Type, XML Data Type, SQLLoader, Data Pump, Stored Procedures, Event Scheduling, Query Optimization Techniques, SQL Profiling.

e. Textbooks / Reference Books

1. Alan Beaulieu, "Learning SQL: Generate, Manipulate, and Retrieve Data", O'Reilly Media, 2020.
2. Itzik Ben-Gan, "T-SQL Fundamentals", Microsoft Press, 2016.
3. Joe Celko, "Joe Celko's SQL for Smarties: Advanced SQL Programming", Morgan Kaufmann, 2014.
4. Anthony Molinaro, "SQL Cookbook: Query Solutions and Techniques for Database Developers", O'Reilly Media, 2005.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT713	Web Analytics	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the principles and importance of web analytics in digital marketing and business.
2. To explore different types of web analytics, metrics, and frameworks used in analyzing online user behavior.
3. To develop practical skills in using web analytics tools for evaluating website performance and improving user experience.

b. Course Outcomes

CO1	Define web analytics and explain its role in improving digital marketing strategies.
CO2	Conduct qualitative and quantitative analyses of web data to inform business decisions.
CO3	Utilize different types of web metrics and dashboards for monitoring website performance.
CO4	Implement conversion funnels and analyze data sources to improve website conversion rates.
CO5	Apply emerging analytics techniques in e-commerce, mobile, and social media analytics.

c. Syllabus

Module -1	Web Analytics Fundamentals
Chapter 1	Introduction - What is Web Analytics - Importance of Web Analytics - Web Analytics Process - Types of Web Analytics - Web Analytics Technical Requirements - Web Analytics 2.0 Framework - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The evolution of web analytics and its impact on digital marketing strategies.
Chapter 2	Qualitative Analysis - Heuristic evaluations - Conducting a heuristic evaluation - Benefits of heuristic evaluations - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Best practices in conducting heuristic evaluations for website improvement.
Chapter 3	Site Visits and Surveys - Conducting a site visit - Benefits of site visits - Website surveys - Post-visit surveys - Creating and running a survey - Benefits of surveys - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Crafting effective surveys for capturing user feedback.
Chapter 4	Web Metrics - Key metrics - Dashboard - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Utilizing dashboards for real-time web performance monitoring.

Chapter 5	Conversion - Goals - Funnels - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Implementing and optimizing conversion funnels for online businesses.
Chapter 6	Data Sources - Server Log - Visitors Data - Search Engine Statistics and Conversion Funnels - Data Segmentation - Analysis - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The importance of data segmentation in personalized marketing.
Module -2	Advanced Web Analytics
Chapter 7	Emerging Analytics - E-Commerce - Mobile Analytics - A/B testing - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: A/B testing strategies for optimizing user experience.
Chapter 8	Social Media Analytics and Annotation and Reporting - Sentimental Analysis - Text Analysis - Automated - Actionable - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Leveraging social media analytics to understand customer sentiment.
Chapter 9	Web Analytics 2.0 - Introduction to analytic 2.0 - Competitive intelligence analysis - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The impact of competitive intelligence analysis on strategic decision-making.

Chapter 10	CI Data Sources - Toolbar data - Panel data - ISP data - Search engine data - Hybrid data - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Integrating data sources to build a comprehensive web analytics framework.
Module -3	Web Analytics in Practice
Chapter 11	Emerging Analytics - Measuring the New Social Web: The Data Challenges - Analyzing Mobile Customer Experience - Measuring the Success of Blogs - Quantifying the Impact of Twitter - Analyzing Performance of Videos - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Data challenges in measuring customer engagement across different digital platforms.
Chapter 12	Website Traffic Analysis - Comparing long term traffic trends - Analyzing competitive site overlap and opportunities - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Techniques for analyzing website traffic patterns over time.
Chapter 13	Google Analytics - Audience analysis - Acquisition analysis - Conversion analysis - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Utilizing Google Analytics for actionable insights into website traffic and behavior.

Chapter 14	Google Website - Google website optimizer - Implementation technology - Privacy issues - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Ethical considerations in web analytics and privacy protection.
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d. Self-study topics for Advance learners:

- The evolution of web analytics and its impact on digital marketing strategies.
- Best practices in conducting heuristic evaluations for website improvement.
- Crafting effective surveys for capturing user feedback.
- Utilizing dashboards for real-time web performance monitoring.
- Implementing and optimizing conversion funnels for online businesses.
- The importance of data segmentation in personalized marketing.
- A/B testing strategies for optimizing user experience.
- Leveraging social media analytics to understand customer sentiment.
- The impact of competitive intelligence analysis on strategic decision-making.
- Integrating data sources to build a comprehensive web analytics framework.
- Data challenges in measuring customer engagement across different digital platforms.
- Techniques for analyzing website traffic patterns over time.
- Utilizing Google Analytics for actionable insights into website traffic and behavior.
- Ethical considerations in web analytics and privacy protection.

e. Textbooks / Reference Books

- Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity by Avinash Kaushik.
- Google Analytics Breakthrough: From Zero to Business Impact by Feras Alhlou, Shiraz Asif, and Eric Fettman.
- Web Analytics Action Hero: Using Analysis to Gain Insight and Optimize Your Business by Brent Dykes.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT714	Digital Media Analytics	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the concepts and practices involved in digital media analytics.
2. To explore various digital media platforms and the analytical methods to measure their effectiveness.
3. To develop skills in utilizing digital media for business intelligence, marketing, and competitive analysis.

b. Course Outcomes

CO1	Understand digital media analytics and its importance in contemporary digital marketing strategies.
CO2	Apply analytical methods to evaluate the effectiveness and reach of digital media campaigns.
CO3	Design and implement SEO and SEM strategies to enhance digital visibility and business growth
CO4	Utilize social media and other digital platforms for targeted marketing and audience engagement.
CO5	Analyze and manage online reputation and ecommerce platforms to optimize digital presence and performance.

c. Syllabus

Module -1	Introduction to Digital Media Analytics & Social Media Research Plans
Chapter 1	What is Digital Media Analytics - What is digital media analytics - Importance - Competitive analysis - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The evolution of digital media analytics in tracking consumer behavior online.
Chapter 2	Working of Digital Media - Digital media in business improvement - Interacting with digital media - How to obtain useful data from digital media - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Strategies for effectively interacting with and extracting valuable data from digital media.
Chapter 3	Website Planning and Creation - Website creation - Planning and organizing - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Best practices in web design and development for maximum user engagement.
Chapter 4	Business Intelligence - Introduction to business intelligence - Role of business intelligence - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The impact of business intelligence on strategic decision-making in digital marketing.
Chapter 5	Search Engine Optimization - Introduction to SEO - Meta tags - Meta descriptions - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Techniques for optimizing web content to enhance search engine ranking.

Chapter 6	Search Engine Marketing - Search engine - Digital marketing - Search Engine Result Page (SERP) - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: The role of SEM in driving targeted traffic to websites.
Chapter 7	Social Media Marketing - Social media - Advertisements and online expansion - Social media campaigns - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Effective social media marketing strategies for brand visibility and engagement.
Module -2	Earned and Paid Media
Chapter 8	Earned Media - Earned media activities - Promotions - Content strategy - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Maximizing the impact of earned media on public relations and marketing.
Chapter 9	Web Analytics - Digital media planning - Digital media purchasing - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Utilizing web analytics to tailor digital media plans and purchase strategies.
Chapter 10	Email Marketing - Email marketing - Target groups - Email campaign analytics - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Effective email marketing techniques and metrics for performance analysis.

Chapter 11	Mobile Marketing - App installation - App metrics - App Store Optimization (ASO) - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Strategies for enhancing mobile app visibility and user engagement through ASO.
Module -3	Digital Media Management
Chapter 12	Ecommerce Management - Ecommerce store - Development, management of platform - Online marketplace - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Key strategies for managing and optimizing ecommerce operations.
Chapter 13	Online Reputation Management - ORM - AdSense - Blogging - Affiliate marketing - Summary and self-assessment questions - References/Reference Reading - SELF STUDY TOPIC: Techniques for managing online reputation and monetizing digital content.
Chapter 14	Digital Media Reports and Audits - Facebook, Instagram, Twitter - Social media reporting - Social media auditing - Summary and self-assessment questions - References/Reference Reading* - SELF STUDY TOPIC: Methods for conducting social media audits and creating reports to assess campaign effectiveness.

d. Self-study topics for Advance learners:

- The evolution of digital media analytics in tracking consumer behavior online.
- Strategies for effectively interacting with and extracting valuable data from digital media.
- Best practices in web design and development for maximum user engagement.
- The impact of business intelligence on strategic decision-making in digital marketing.
- Techniques for optimizing web content to enhance search engine ranking.
- The role of SEM in driving targeted traffic to websites.

- Effective social media marketing strategies for brand visibility and engagement.
- Maximizing the impact of earned media on public relations and marketing.
- Utilizing web analytics to tailor digital media plans and purchase strategies.
- Effective email marketing techniques and metrics for performance analysis.
- Strategies for enhancing mobile app visibility and user engagement through ASO.
- Key strategies for managing and optimizing ecommerce operations.
- Techniques for managing online reputation and monetizing digital content.
- Methods for conducting social media audits and creating reports to assess campaign effectiveness.

e. Textbooks / Reference Books

1. **Web Analytics 2.0** by Avinash Kaushik.
2. **Social Media ROI: Managing and Measuring Social Media Efforts in Your Organization** by Olivier Blanchard.
3. **Digital Marketing Analytics: Making Sense of Consumer Data in a Digital World** by Chuck Hemann and Ken Burbary.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT715	IOT and Data Analytics	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the concepts, components, and applications of the Internet of Things (IoT).
2. To explore the relevance and impact of IoT on various industries and everyday life.
3. To learn about IoT standards, protocols, and device management analytics.
4. To apply statistical analysis, machine learning, and time series modeling techniques to IoT data.

b. Course Outcomes

CO1	Understand the fundamentals, pillars, and components of IoT systems.
CO2	Identify the applications and challenges of IoT implementation in various domains.
CO3	Analyze sensor data using statistical methods and machine learning techniques.
CO4	Apply device management analytics and communication protocols in IoT networks.
CO5	Evaluate the relevance and impact of IoT on future developments and everyday life.

c. Syllabus

Module -1	*Introduction to Internet of Things (IoT)*
Chapter 1	Internet of Things (IoT) - Introduction to IoT - Definitions of IoT - Pillars of IoT - Examples of IoT - IoT Standards - Key Standards for IEEE activities - Components of IoT System - Relevance of IoT for future - Improvements in future with IoT - Challenges - Applications - Summary - Self-Assessment Questions - Self-Study Topics: IoT Architecture, IoT Ecosystem
Chapter 2	IoT Concepts - Introduction - IoT Evaluation Technology - IoT and SCADA - IoT and M2M - IoT and Big Data - Summary - Self-Assessment Questions - Self-Study Topics: IoT and Cloud Computing, IoT and Edge Computing
Chapter 3	IoT Standards - Requirement of international standards - IoT standards in practice - Operating platforms - Summary - Self-Assessment Questions - References - Self-Study Topics: IoT Communication Protocols, IoT Security
Chapter 4	Components of IoT System - Introduction - IoT System Design - Prototype Development - Summary - Self-Assessment Questions - Self-Study Topics: IoT Sensors and Actuators, IoT Gateways

Chapter 5	Relevance of IoT System for Future - IoT in Everyday Life - IoE (Internet of Everything) - IoT and Privacy - Summary - Self-Assessment Questions - Self-Study Topics: IoT and Blockchain, IoT and AI
Chapter 6	IoT Applications - Introduction - History of IoT - Applications of IoT - IoT for Smart Cities - IoT use cases for smart cities - IoT based Smart Parking Systems - IoT based Smart Water Management Technology - IoT Applications in Water Management - IoT based smart traffic management system - Summary - Self-Assessment Questions - Self-Study Topics: IoT in Healthcare, IoT in Agriculture
Module -2	IoT in India and Challenges
Chapter 7	IoT in India - Introduction - IoT and India - Definition & Vision - Pillars of IoT - Applications of IoT in government - Role of IoT in FASTag - IoT Based Ration Distribution System Using Aadhar Card - IoT in the Banking and Financial Services - Role of IoT In Rural Development - Challenges in Securing IoT - Leading IoT initiatives in India - Summary - Self-Assessment Questions - Self-Study Topics: IoT Start-ups in India, IoT and Digital India
Chapter 8	Challenges in IoT Implementation - Critical Applications - Big Data Management - Connectivity Challenges - Summary - Self-Assessment Questions - Self-Study Topics: IoT and Cybersecurity, IoT and Data Privacy

Chapter 9	Sensor Analytics - Types of Sensors - Data Capturing - Data Handling and Preprocessing - Data Conversion - Time and Frequency Analysis - Selection and Cleaning - Edge Analytics - Summary - Self-Assessment Questions - Self-Study Topics: IoT Sensor Fusion, IoT Sensor Calibration
Chapter 10	Device Management Analytics - Interoperability in IoT - Sensor Networks - Communication Protocols - SDN for IoT - Cloud Computing - Summary - Self-Assessment Questions - Self-Study Topics: IoT Device Management Platforms, IoT Fleet Management
Module -3	Data Analytics for IoT
Chapter 11	Statistical Analysis - Exploring Data - Multiple Linear Regression - Using Regression on IoT Data - Correlation - Summary - Self-Assessment Questions - Self-Study Topics: IoT Data Visualization, IoT Data Mining
Chapter 12	Single Value Decomposition - Introduction - Singular Value Decomposition - Numerical Procedure - Applications - Data Reduction - Applications of the SVD - Summary - Self-Assessment Questions - Self-Study Topics: SVD for IoT Data Compression, SVD for IoT Data Denoising

Chapter 13	Machine Learning - Introduction - Information Bases - Machine Learning Tools - Summary - Self-Assessment Questions - Self-Study Topics: Deep Learning for IoT, Reinforcement Learning for IoT
Chapter 14	Time Series Modeling - Introduction - Implementation on IoT Data - Summary - Self-Assessment Questions - References - Self-Study Topics: ARIMA Models for IoT Data, LSTM Models for IoT Data

d. Self-study topics for Advance learners:

- IoT Architecture, IoT Ecosystem
- IoT and Cloud Computing
- IoT and Edge Computing,
- IoT Communication Protocols
- IoT Security Standard,
- IoT Sensors and Actuators
- IoT Gateways
- IoT and Blockchain
- IoT and AI
- IoT in Healthcare
- IoT in Agriculture
- IoT Start-ups in India
- IoT and Digital India
- IoT and Cybersecurity
- IoT and Data Privacy
- IoT Sensor Fusion
- IoT Sensor Calibration
- IoT Device Management Platforms
- IoT Fleet Management
- IoT Data Visualization
- IoT Data Mining
- SVD for IoT Data Compression
- SVD for IoT Data Denoising
- Deep Learning for IoT
- Reinforcement Learning for IoT
- ARIMA Models for IoT Data, LSTM Models for IoT Data

e. Textbooks / Reference Books

1. Vijay Madiseti and Arshdeep Bahga, "Internet of Things: A Hands-On Approach", VPT, 2014.
2. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", Cisco Press, 2017.
3. Arshdeep Bahga, Vijay Madiseti, "Internet of Things: A Hands-On Approach", Universities Press, 2015.
4. Rajkumar Buyya, Amir Vahid Dastjerdi, "Internet of Things: Principles and Paradigms", Morgan Kaufmann, 2016.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus: Semester III - Artificial Intelligence and Machine Learning

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT716	Machine Learning in Python	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamental concepts and techniques of machine learning.
2. To learn how to preprocess and analyze data for machine learning tasks.
3. To explore various machine learning algorithms for classification, regression, and clustering.
4. To apply machine learning techniques to real-world problems using Python and scikit-learn library.

b. Course Outcomes

CO1	Understand the basic terminology, notations, and types of machine learning.
CO2	Preprocess and transform raw data into suitable formats for machine learning algorithms.
CO3	Implement and evaluate various machine learning algorithms for classification, regression, and clustering tasks.
CO4	Apply dimensionality reduction techniques to compress and visualize high-dimensional data.
CO5	Utilize ensemble learning methods to improve the performance of machine learning models.

c. Syllabus

Module -1	Introduction to Machine Learning and Data Preprocessing
Chapter 1	Machine Learning and Classification Algorithms - Building intelligent machines to transform data into knowledge - Types of machine learning - An introduction to the basic terminology and notations - A roadmap for building machine learning systems - Using Python for machine learning - Artificial neurons – a brief glimpse into the early history of machine learning - Implementing a perceptron learning algorithm in Python - Adaptive linear neurons and the convergence of learning - Summary - Self Assessment Questions - Self-Study Topics: Bias-Variance Tradeoff, Regularization Techniques
Chapter 2	A Tour of Machine Learning Classifiers Using Scikit-learn and Data Preprocessing - Choosing a classification algorithm - Modeling class probabilities via logistic regression - Maximum margin classification with support vector machines - Solving nonlinear problems using a kernel SVM - Decision tree learning - K-nearest neighbors – a lazy learning algorithm - Building Good Training Sets – Data Preprocessing - Handling categorical data - Partitioning a dataset in training and test sets - Bringing features onto the same scale - Selecting meaningful features - Summary - Self Assessment Questions - Self-Study Topics: Cross-Validation, Hyperparameter Tuning
Module -2	Dimensionality Reduction and Ensemble Learning
Chapter 3	Compressing Data via Dimensionality Reduction - Unsupervised dimensionality reduction via principal component analysis - Principal component analysis in scikit-learn - Supervised data compression via linear discriminant analysis - Using kernel principal component analysis for nonlinear mappings - Kernel principal component analysis in scikit-learn - Summary - Self Assessment Questions - Self-Study Topics: t-SNE, Autoencoders

Chapter 4	Combining Different Models for Ensemble Learning - Learning with ensembles - Implementing a simple majority vote classifier - Combining different algorithms for classification with majority vote - Evaluating and tuning the ensemble classifier - Bagging – building an ensemble of classifiers from - Leveraging weak learners via adaptive boosting - Summary - Self Assessment Questions - Self-Study Topics: Stacking, Gradient Boosting Machines
Module -3	Sentiment Analysis, Regression, and Clustering
Chapter 5	Applying Machine Learning to Sentiment Analysis - Obtaining the IMDb movie review dataset - Introducing the bag-of-words model - Assessing word relevancy via term frequency-inverse - Cleaning text data - Processing documents into tokens - Training a logistic regression model for document classification - Working with bigger data – online algorithms and out-of-core learning - Summary - Self Assessment Questions - Self-Study Topics: Word Embeddings, Recurrent Neural Networks for Sentiment Analysis
Chapter 6	Predicting Continuous Target Variables with Regression Analysis - Introducing a simple linear regression model - Exploring the Housing Dataset - Visualizing the important characteristics of a dataset - Implementing an ordinary least squares linear regression model - Estimating the coefficient of a regression model via scikit-learn - Evaluating the performance of linear regression models - Turning a linear regression model into a curve – polynomial regression - Modeling nonlinear relationships in the Housing Dataset - Dealing with nonlinear relationships using random forests - Summary - Self Assessment Questions - Self-Study Topics: Regularized Regression (Ridge, Lasso), Gaussian Process Regression

Chapter 7	Working with Unlabeled Data – Clustering Analysis - Grouping objects by similarity using k-means - K-means++ - Hard versus soft clustering - Using the elbow method to find the optimal number of clusters - Quantifying the quality of clustering via silhouette plots - Organizing clusters as a hierarchical tree - Applying agglomerative clustering via scikit-learn - Summary - Self Assessment Questions - Self-Study Topics: DBSCAN, Gaussian Mixture Models
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d. Self-study topics for Advance learners:

- Bias-Variance Tradeoff, Regularization Techniques, Cross-Validation, Hyperparameter Tuning, t-SNE, Autoencoders, Stacking, Gradient Boosting Machines, Word Embeddings, Recurrent Neural Networks for Sentiment Analysis, Regularized Regression (Ridge, Lasso), Gaussian Process Regression, DBSCAN, Gaussian Mixture Models.

e. Textbooks / Reference Books

1. Sebastian Raschka, Vahid Mirjalili, "Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2", Packt Publishing, 2019.
2. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems", O'Reilly Media, 2019.
3. Andreas C. Müller, Sarah Guido, "Introduction to Machine Learning with Python: A Guide for Data Scientists", O'Reilly Media, 2016.
4. Giuseppe Bonaccorso, "Mastering Machine Learning Algorithms: Expert Techniques to Implement Popular Machine Learning Algorithms and Fine-Tune Your Models", Packt Publishing, 2018.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT717	Statistics and Python in Machine Learning	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the Python programming language and its applications in data analysis and machine learning.
2. To develop an understanding of statistical concepts and their implementation in Python.
3. To learn data manipulation techniques using NumPy and Pandas libraries.
4. To explore various mathematical and statistical functions in Python.
5. To gain knowledge of data visualization techniques and libraries in Python.

b. Course Outcomes

CO1	Understand the fundamentals of Python programming language and its applications.
CO2	Implement file handling, exception handling, and functional programming concepts in Python.
CO3	Manipulate and analyze data using NumPy and Pandas libraries.
CO4	Apply statistical concepts and probability distributions for data analysis and modeling.
CO5	Utilize mathematical and statistical functions in Python for data analysis tasks.
CO6	Visualize data effectively using various plotting techniques and libraries in Python.

c. Syllabus

Module -1	Introduction to Python and Programming Concepts
Chapter 1	Getting Started With Python - Introduction to Python - Python Programming Domains - Python Installation - Setting Path in Python - Execute Modes in Python - Python Keywords, Identifiers, and Variables - Python Statement - Structuring with Indentation - Comments in Python - Python Data Types - Python Lists - Python Tuples - Shallow and Deep Copy - Python Dictionaries - Python Sets and Types - Python Operators - Python Control Structures - Summary - Self-Assessment Question SELF STUDY TOPIC Object-Oriented Programming in Python
Chapter 2	Functions, File Handling and Exceptions - Python Functions - Definition of Recursion - Python Input and Output - Python File Handling - With Statement - Exception Handling - User-defined Exceptions - Built-in Exceptions - Summary - Self-Assessment Questions SELF STUDY TOPIC Regular Expressions in Python
Module -2	Data Manipulation and Statistical Concepts

Chapter 3	Data Manipulation Using NumPy and Pandas - Python Libraries - NumPy: Arrays and Matrices - Pandas: Data Manipulation - File I/O - The Fisher's F-distribution - Case Study - Summary - Self-Assessment Questions SELF STUDY TOPIC Advanced Pandas Techniques
Chapter 4	Introduction to Statistics and Probability - Statistics in Python - Probability Distributions - 9 Most Commonly Used Probability Distributions - Hypothesis Testing (T-Test) - Linear Regression - Statistical Data Modeling - Introduction to Bayesian Thinking - Markov Chains in Python - Summary - Self-Assessment Questions SELF STUDY TOPIC Bayesian Statistics
Module -3	Mathematical Functions, Statistics, and Data Visualization
Chapter 5	Mathematical and Statistics Functions - Mathematical Functions in Python - Statistical Functions in Python - Histogram (hist) Function with Multiple Data Sets - Box Plot vs. Violin Plot Comparison - Producing Multiple Histograms Side by Side - Barchart - Scatter Plots - Line-style Reference - Scatter Hist - Pie Chart - Polar Demo - Summary - Self-Assessment Questions SELF STUDY TOPIC Advanced Mathematical Functions in Python

Chapter 6	Data Visualization in Python -Introduction to Data Visualizatio - Advanced Visualization Methods - Visualization Libraries in Python - Plots in Python - Colormap in Plots - Style Sheets Reference - Summary - Self-Assessment Questions SELF STUDY TOPIC Interactive Data Visualization

d. Self-study topics for Advance learners:

- Object-Oriented Programming in Python
- Regular Expressions in Python
- Advanced Pandas Techniques
- Bayesian Statistics
- Advanced Mathematical Functions in Python
- Interactive Data Visualization

e. Textbooks / Reference Books

1. Wes McKinney, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython," O'Reilly Media, 2017.
2. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems," O'Reilly Media, 2019.
3. Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data," O'Reilly Media, 2016.
4. Allen B. Downey, "Think Stats: Exploratory Data Analysis in Python," O'Reilly Media, 2014.
5. Cyrille Rossant, "IPython Interactive Computing and Visualization Cookbook: Over 100 hands-on recipes to sharpen your skills in high-performance numerical computing and data science in the Jupyter Notebook," Packt Publishing, 2018

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT718	Business Application of Machine Learning	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the basic concepts of artificial intelligence and machine learning.
2. To explore the applications of machine learning in various business domains, including customer service, hospitality, banking, and retail.
3. To gain insights into the future potential and technological advancements in machine learning.
4. To develop a comprehensive understanding of how machine learning can be leveraged to solve real-world business problems.

b. Course Outcomes

CO1	Understand the fundamental concepts of artificial intelligence and machine learning and their relationship with big data and data science.
CO2	Identify and analyze the applications of machine learning in customer service, such as chatbots and call center automation.
CO3	Evaluate the use of machine learning techniques in the hospitality industry for dynamic pricing and demand forecasting.
CO4	Assess the impact of machine learning in banking and financial services, including robo-advisers and loan/insurance underwriting.
CO5	Examine the role of machine learning in retail, focusing on online recommendation systems and inventory optimization.

c. Syllabus

Module -1	Introduction to AI and ML in Customer Service
Chapter 1	Introduction to Artificial Intelligence & Machine Learning - Artificial Intelligence: Basic Concepts - AI, Big Data, Data Science, Machine Learning: Relationship - Beyond the AI Hype - Summary - Self-Assessment Questions - Self-Study Topics: Ethics in AI, AI and Job Market
Chapter 2	Machine Learning Application in Customer Service - Customer Experience - Chatbots - Call Center Automation - Summary - Self-Assessment Questions - Self-Study Topics: Sentiment Analysis, Personalization in Customer Service
Module -2	ML in Hospitality and Banking & Financial Services
Chapter 3	Machine Learning Application in Hospitality - Best Fit / Dynamic Pricing - Demand Forecasting - Summary - Self-Assessment Questions - Self-Study Topics: Revenue Management, Customer Segmentation in Hospitality
Chapter 4	Machine Learning Application in Banking & Financial Services - Robo-Advisers based Portfolio Management - Loan & Insurance Underwriting - Summary - Self-Assessment Questions - Self-Study Topics: Fraud Detection, Risk Assessment
Module -3	ML in Retail and Future Potential
Chapter 5	Machine Learning Application in Retail - Online Recommendation - Inventory Optimization - Summary - Self-Assessment Questions - Self-Study Topics: Customer Churn Prediction, Market Basket Analysis
Chapter 6	Future Potential of Machine Learning - Technological Advances in Machine Learning - Future Possibilities Unlimited - Summary - Self-Assessment Questions - Self-Study Topics: Explainable AI, Quantum Machine Learning

Chapter 7	Summary - Machine Learning Application Recap - Self Assessment Questions
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d. Self-study topics for Advance learners:

- Ethics in AI, AI and Job Market, Sentiment Analysis, Personalization in Customer Service, Revenue Management, Customer Segmentation in Hospitality, Fraud Detection, Risk Assessment, Customer Churn Prediction, Market Basket Analysis, Explainable AI, Quantum Machine Learning.

e. Textbooks / Reference Books

1. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems", O'Reilly Media, 2019.
2. Tom M. Mitchell, "Machine Learning", McGraw-Hill Education, 1997.
3. Bing Liu, "Sentiment Analysis: Mining Opinions, Sentiments, and Emotions", Cambridge University Press, 2015.
4. Francesco Ricci, Lior Rokach, Bracha Shapira, "Recommender Systems Handbook", Springer, 2015.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT719	Deep Learning and NLP	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of artificial intelligence, machine learning, deep learning, and natural language processing.
2. To review and apply essential mathematical concepts in the context of deep learning.
3. To gain knowledge of various deep learning algorithms, platforms, and frameworks.
4. To develop practical skills in implementing deep learning projects and applying NLP techniques.

b. Course Outcomes

CO1	Understand the core concepts of artificial intelligence, machine learning, deep learning, and natural language processing.
CO2	Apply linear algebra, probability, statistics, and calculus concepts in the context of deep learning.
CO3	Comprehend the structure and functioning of artificial neural networks and model neurons.
CO4	Implement various deep learning algorithms using popular platforms and frameworks.
CO5	Develop practical skills in building deep learning projects and applying NLP techniques to real-world problems.

c. Syllabus

Module -1	Fundamentals of AI, Deep Learning, and NLP
Chapter 1	Artificial Intelligence Overview - Artificial Intelligence - Machine Learning - Deep Learning - Natural Language Processing - Summary - Self-Assessment Questions - Self-Study Topics: History of AI, Ethical Considerations in AI
Chapter 2	Review of Mathematical Concepts - Overview - Linear Algebra - Probability & Statistics - Calculus - Summary - Self-Assessment Questions - Self-Study Topics: Matrix Calculus, Information Theory
Chapter 3	Introduction to Artificial Neural Networks - Context - Research into ANN - Brain & Neurons - Initial concepts in ANN - Summary - Self-Assessment Questions - Self-Study Topics: Biological Neural Networks, Neuromorphic Computing
Module -2	Deep Learning Algorithms, Platforms, and Projects
Chapter 4	Modelling a Neuron - Network Topology - Neural Network Learning - Activation Functions - Summary - Self-Assessment Questions - Self-Study Topics: Gradient Descent Optimization, Regularization Techniques

Chapter 5	Deep Learning Algorithms - Perceptron Learning Algorithm - Back Propagation Neural Network Algorithm - Deep Boltzmann Machine (DBM) Learning Algorithm - Deep Belief Network (DBN) Learning Algorithm - Convolutional Neural Network (CNN) Learning Algorithm - Stacked Autoencoder Learning Algorithm - Summary - Self-Assessment Questions - Self-Study Topics: Generative Adversarial Networks (GANs), Recurrent Neural Networks (RNNs)
Chapter 6	Deep Learning Platforms and Frameworks - Overview - TensorFlow - Keras - Microsoft Cognitive Toolkit (CNTK) - PyTorch - Others - Summary - Self-Assessment Questions - Self-Study Topics: MXNet, Caffe2
Chapter 7	Deep Learning Project - API Environment Overview - Project Scenario and Solution - Summary - Self-Assessment Questions - Self-Study Topics: Transfer Learning, Hyperparameter Tuning
Module -3	Natural Language Processing
Chapter 8	NLP Introduction - Overview and History of NLP - Business Applications of NLP - Summary - Self-Assessment Questions - Self-Study Topics: Computational Linguistics, Sentiment Analysis
Chapter 9	Approach to NLP - Introduction to NLP and NLTK - Text Pre-processing - Feature Engineering on Text - NLP use cases - Summary - Self-Assessment Questions - Self-Study Topics: Word Embeddings, Transformer Models (BERT,

d. Self-study topics for Advance learners:

History of AI, Ethical Considerations in AI, Matrix Calculus, Information Theory, Biological Neural Networks, Neuromorphic Computing, Gradient Descent Optimization, Regularization Techniques, Generative Adversarial Networks (GANs), Recurrent Neural Networks (RNNs), MXNet, Caffe2, Transfer Learning, Hyperparameter Tuning, Computational Linguistics, Sentiment Analysis, Word Embeddings, Transformer Models (BERT, GPT).

e. Textbooks / Reference Books

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", MIT Press, 2016.
2. Francois Chollet, "Deep Learning with Python", Manning Publications, 2017.
3. Steven Bird, Ewan Klein, Edward Loper, "Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit", O'Reilly Media, 2009.
4. Yaser Abu-Mostafa, Malik Magdon-Ismail, Hsuan-Tien Lin, "Learning From Data", AMLBook, 2012.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT720	Web, Social Analytics and Visualization	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals, importance, and applications of web analytics and social media analytics.
2. To explore various web analytics tools and techniques for analyzing website traffic, user behavior, and performance.
3. To gain insights into social media platforms, analytics tools, and metrics for effective social media strategy.
4. To learn about social media ROI, risks, benchmarking, reporting, and data visualization techniques.

b. Course Outcomes

CO1	Understand the concepts, benefits, and challenges of web analytics and social media analytics.
CO2	Apply various web analytics tools and techniques to analyze website traffic, user behavior, and performance.
CO3	Utilize social media analytics tools and metrics to develop effective social media strategies.
CO4	Evaluate social media ROI, assess risks, and optimize social media presence.
CO5	Create meaningful reports and visualizations of web and social media analytics data.

c. Syllabus

Module -1	Introduction to Web Analytics and Tools
Chapter 1	Introduction to Web Analytics - Web analytics: introduction - The purpose of web analytics - The chronological journey of web analytics - Benefits of web analytics - Challenges of web analytics - Importance of web analytics - Unlock your website's potential through web analytics - Grow your business with web analytics - Using web analytics basics - The five 'whs' of web analytics - Most commonly used terms - Basic criteria's to choose a web analytic tool - Top 7 web analytics tools - Categories of web analytics software - Web analytic dashboard - Metrics categories - Site referrers - Identifying your audience - How to identify the most important pages? - Key performance indicators - Ten most common web analytics mistakes and pitfalls - Web analytics: best practices - How to know if your analytic data is performing well. - Minimizing cost using testing software - Web analytics methods - Summary - Self assessment questions - Self-Study Topics: A/B Testing, Heat Maps

Chapter 2	Web Analytics Tools - Web analytics: objectives and process - Web analytics tools - Traffic & trends of websites - Marketing automation & conversion metrics - Keywords, SEO, PPC & competitor research - User behaviour & flow - Google analytics - Analyzing data through google analytics - Optimizely - Kissmetrics - Crazy egg - Data source - Dashboard - Goals of analytics - Web traffic data analysis - The top 5 pillars of web analytics - Increasing your site's visibility through web analytics - Summary - Self assessment questions - Self-Study Topics: Adobe Analytics, Mixpanel
Module -2	Introduction to Social Media Analytics and Platforms
Chapter 3	Introduction to Social Media Analytics - Social media - From Social Media to Social Media Analytics - Social media analytics - Importance of tracking Social Media Analytics - Benefits of social media - Social media communities - Data analytics - Data analytics in social media - Tools for social media analytics - Structured and Unstructured Data - How social-media analytics help individuals and organizations - How to use social data analytics to improve your marketing strategies - Popular Social Media Tools and Platforms - Benefits of adopting Social Media Strategy - Organizational Goals for Social Media - Social Media Impact on Business - Summary - Self assessment questions - Self-Study Topics: Sentiment Analysis, Influencer Marketing

Chapter 4	Social Media Platforms and Analytics - Social Media Platform - Facebook Insights - Twitter Analytics - Pinterest Analytics - LinkedIn Analytics - Google+ Analytics - Instagram Analytics - YouTube Analytics - Blogging Analytics - How to Turn Social Media Analytical Data Into Actionable Insights - Gaining an Advantage from Social Media Trends - Summary - Self assessment questions - Self-Study Topics: TikTok Analytics, Snapchat Analytics
Module -3	Social Media Metrics, ROI, Risks, and Visualization
Chapter 5	Social Media Metrics, ROI and Risks - Social Media Metrics: Introduction - Tools to track Social media metric - Social media ROI (Return on Investment) - Importance of measuring ROI - Social media ROI: objectives, goals, and metrics - Social media ROI tools - Report your social media ROI - Big Data Analytics on Social Media - Social Media Mining - Issues in Mining Social Media - Social Media Optimization & Search Engine Optimization - Strategies for Social Media Optimization - Social Media risks - Social Media Risk Assessment - Types of Business Risks - Steps to Effectively Managing Social Media Risk - Legal & Regulatory Challenges with Social Media - Summary - Self assessment questions - Self-Study Topics: Crisis Management, Reputation Management

Chapter 6	Social Media Benchmarking, Reporting And Visualization - Improving performance through Social Media Benchmarking - Metrics to Benchmark - Things can be benchmarked on social media - Social media analytics reports - Visualize your Social Media Analytics - Tips for visualizing social media data - Best Data Visualization Tools - Summary - Self assessment questions - Self-Study Topics: Interactive Dashboards, Infographics
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d. Self-study topics for Advance learners:

- A/B Testing, Heat Maps, Adobe Analytics, Mixpanel, Sentiment Analysis, Influencer Marketing, TikTok Analytics, Snapchat Analytics, Crisis Management, Reputation Management, Interactive Dashboards, Infographics.

e. Textbooks / Reference Books

1. Avinash Kaushik, "Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity", Sybex, 2009.
2. Matthew A. Russell, "Mining the Social Web: Data Mining Facebook, Twitter, LinkedIn, Instagram, GitHub, and More", O'Reilly Media, 2019.
3. Gohar F. Khan, "Creating Value with Social Media Analytics: Managing, Aligning, and Mining Social Media Text, Networks, Actions, Location, Apps, Hyperlinks, Multimedia, & Search Engines Data", CreateSpace Independent Publishing Platform, 2018.
4. Krista Neher, "Visual Social Marketing For Dummies", For Dummies, 2013.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus: Semester IV - Cloud Computing

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT751	Introduction to Google Cloud Services	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of cloud computing and its benefits.
2. To explore various cloud storage options, including SQL, large-scale SQL, and structured data.
3. To learn about computing resources in the cloud, such as virtual machines and serverless applications.
4. To gain knowledge of machine learning services, data processing, and analytics in the cloud.

b. Course Outcomes

CO1	Understand the concept of cloud computing, its benefits, and the infrastructure of cloud data centers.
CO2	Configure and manage various cloud storage options, including SQL, NewSQL, and Bigtable.
CO3	Deploy and scale virtual machines and serverless applications in the cloud.
CO4	Utilize cloud-based machine learning services for image annotation, speech recognition, and translation.
CO5	Perform large-scale data processing and analytics using BigQuery and Cloud Dataflow.

c. Syllabus

Module -1	Introduction to Cloud and Storage
Chapter 1	Introduction To Cloud - What is Cloud? - Why Cloud & what are its benefits? - Summary - Self-Assessment Questions - Self-Study Topics: Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid)
Chapter 2	Cloud Data Center - Locations of data centers - Fault tolerance & isolation levels - Safety concerns - Performance & resource isolation - Summary - Self-Assessment Questions - Self-Study Topics: Data Center Tiers, Green Data Centers
Chapter 3	Storage: SQL - What is cloud SQL? - Configuring Cloud SQL - Scaling up and down - Backup and restore - Cost - Summary - Self-Assessment Questions - Self-Study Topics: Database Migration, Database Security
Chapter 4	Storage: Large-scale SQL - What is NewSQL and Spanner? - Interaction to Cloud spanner - Some advanced concepts - Why to use Cloud Spanner? - Summary - Self-Assessment Questions - Self-Study Topics: ACID Properties, Distributed Transactions
Module -2	Storage, Computing, and Serverless Applications
Chapter 5	Storage: Large Scale Structured Data - What is meant by Bigtable? - Interaction with Cloud Bigtable - Difference between Bigtable & HBase - Case study - Summary - Self-Assessment Questions - Self-Study Topics: Column-Family Databases, NoSQL Databases

Chapter 6	Computing: Virtual Machines - Launching First VM - Dynamic Resources & Load Balancing - Cloud CDN - Why to use GCE? - Summary - Self-Assessment Questions - Self-Study Topics: Containers, Kubernetes
Chapter 7	Serverless Applications - Microservices - What are Google cloud functions? - Interaction with cloud functions - Cost - Summary - Self-Assessment Questions - Self-Study Topics: Event-Driven Architecture, Function-as-a-Service (FaaS)
Chapter 8	Managed DNS Hosting with Google cloud - What is Cloud DNS? - Interaction with Cloud DNS. - Case study - Summary - Self-Assessment Questions - Self-Study Topics: DNS Record Types, DNS Security
Module -3	Machine Learning, Data Processing, and Analytics
Chapter 9	Machine Learning: Cloud Vision & Natural language - Annotating images - How does the natural language API work? - Entity recognition & syntax analysis - Case study - Summary - Self-Assessment Questions - Self-Study Topics: Sentiment Analysis, Named Entity Recognition
Chapter 10	Cloud Speech & Multi Language Translation - Simple Speech Recognition - Continuous Speech Recognition - How does the translation API work? - Language & Text Detection - Summary - Self-Assessment Questions - Self-Study Topics: Speech Synthesis, Language Identification

Chapter 11	Managed Machine Learning With Google Cloud - What is a cloud machine learning Engine? - Interaction with Cloud based machine learning engine. - Summary - Self-Assessment Questions - Self-Study Topics: AutoML, TensorFlow
Chapter 12	Data Processing & Analytics - What is BigQuery? - First Interaction with BigQuery - Cost calculation - Summary - Self-Assessment Questions - Self-Study Topics: Data Warehousing, ETL (Extract, Transform, Load)
Chapter 13	Large-Scale Data processing - Introduction to Apache beam - Cloud dataflow - Interaction with Cloud Dataflow - Summary - Self-Assessment Questions - Self-Study Topics: Apache Spark, MapReduce
Chapter 14	Google Cloud: Managed Event Publishing - What is Cloud pub/sub? - Advanced Concepts - Messaging Patterns - More on Google Cloud Platform/Services - Summary - Self-Assessment Questions - Self-Study Topics: Message Queues, Streaming Analytics

d. Self-study topics for Advance learners:

- Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid), Data Center Tiers, Green Data Centers, Database Migration, Database Security, ACID Properties, Distributed Transactions, Column-Family Databases, NoSQL Databases, Containers, Kubernetes, Event-Driven Architecture, Function-as-a-Service (FaaS), DNS Record Types, DNS Security, Sentiment Analysis, Named Entity Recognition, Speech Synthesis, Language Identification, AutoML, TensorFlow, Data Warehousing, ETL (Extract, Transform, Load), Apache Spark, MapReduce, Message Queues, Streaming Analytics.

e. Textbooks / Reference Books

1. Dan C. Marinescu, "Cloud Computing: Theory and Practice", Morgan Kaufmann, 2017.
2. Tom White, "Hadoop: The Definitive Guide", O'Reilly Media, 2015.
3. Valliappa Lakshmanan, "Data Science on the Google Cloud Platform", O'Reilly Media, 2018.

4. Kai Waehner, "Real-Time Analytics: Techniques to Analyze and Visualize Streaming Data", Wiley, 2014.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT752	Introduction to IBM Cloud Services	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of cloud computing, its history, infrastructure, and workloads.
2. To explore IBM Platform Load Sharing Facilities (LSF) and IBM Platform Symphony for technical cloud computing.
3. To learn about IBM Platform Cluster Manager - Advanced Edition (PCM-AE) and IBM General Parallel File System (GPFS) for technical cloud computing.
4. To gain knowledge of cloud solutions for engineering, life sciences, financial services, and business analytics workloads.

b. Course Outcomes

CO1	Understand the basics of cloud computing, its history, infrastructure, and workloads.
CO2	Apply IBM Platform LSF and IBM Platform Symphony for managing technical cloud computing workloads.
CO3	Utilize IBM PCM-AE and GPFS for managing and optimizing technical cloud computing resources.
CO4	Develop cloud solutions for engineering, life sciences, financial services, and business analytics workloads.
CO5	Implement effective security, governance, and reliability strategies for cloud workloads and services.

c. Syllabus

Module -1	Introduction to Cloud Computing and IBM Platform Solutions
Chapter 1	Introduction - History - Infrastructure - Workloads - Why use Clouds? - Types of Clouds - Summary - Self-Assessment Questions - Self-Study Topics: Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid)
Chapter 2	Understanding Cloud Fundamentals and the Cloud Continuum - Discovering Cloud Basics - Foundational Cloud Delivery Services - Core Cloud Capabilities - Understanding the Cloud Continuum - Summary - Self-Assessment Questions - Self-Study Topics: Cloud Migration Strategies, Cloud Governance
Chapter 3	IBM Platform Load Sharing Facilities for Technical Cloud Computing - Overview - IBM Platform LSF family features and benefits - IBM Platform LSF job management - Resource management - MultiCluster - Summary - Self-Assessment Questions - Self-Study Topics: High-Performance Computing (HPC), Grid Computing
Chapter 4	IBM Platform Symphony for Technical Cloud Computing - Overview - Supported workload patterns - Workload submission - Advanced resource sharing - Dynamic growth and shrinking - Data management - Advantages of Platform Symphony - Summary - Self-Assessment Questions - Self-Study Topics: Distributed Computing, Service-Oriented Architecture (SOA)
Module -2	IBM Platform Solutions and Cloud Solutions for Various Workloads

Chapter 5	IBM Platform Symphony MapReduce - Overview - Key advantages for Platform Symphony MapReduce - Key benefits - Summary - Self-Assessment Questions - Self-Study Topics: Hadoop Ecosystem, Big Data Analytics
Chapter 6	IBM Platform Cluster Manager - Advanced Edition (PCM-AE) for Technical Cloud Computing - Overview - Platform Cluster Manager - Advanced Edition capabilities and benefits - Architecture and components - PCM-AE managed clouds support - PCM-AE: a cloud-oriented perspective - Summary - Self-Assessment Questions - Self-Study Topics: Cluster Orchestration, Virtualization
Chapter 7	The IBM General Parallel File System for Technical Cloud Computing - Overview - GPFS layouts for technical computing - Integration with IBM Platform Computing products - GPFS features for Technical Computing - Summary - Self-Assessment Questions - Self-Study Topics: Parallel I/O, Distributed File Systems
Chapter 8	Solution for Engineering Workloads - Solution overview - Architecture - Components - Use cases - Summary - Self-Assessment Questions - Self-Study Topics: Computer-Aided Design (CAD), Computer-Aided Engineering (CAE)
Module -3	Cloud Solutions for Various Workloads and Cloud Management
Chapter 9	Solution for Life Sciences Workloads - Overview - Architecture - Use cases - Summary - Self-Assessment Questions - Self-Study Topics: Bioinformatics, Drug Discovery

Chapter 10	Solution for Financial Services Workloads - Overview - Architecture - Use cases - Third-party integrated solutions - Summary - Self-Assessment Questions - Self-Study Topics: Risk Analytics, Algorithmic Trading
Chapter 11	Digging Deeper into IaaS and PaaS - Diving into Infrastructure as a Service - Listing the characteristics of IaaS - Exploring PaaS Exploring PaaS - Having the Correct Requirements for IaaS and PaaS - Summary - Self-Assessment Questions - Self-Study Topics: Serverless Computing, Containers
Chapter 12	Managing Cloud Workloads and Services - Understanding Workloads - Use Cases - Looking at the Principles of Workload Management - Seeing Workload Management in a Hybrid Cloud - Connecting Workloads in the Cloud - Managing and Monitoring Workloads - Summary - Self-Assessment Questions - Self-Study Topics: Service Level Agreements (SLAs), Cloud Monitoring Tools
Chapter 13	Improving Security, Governance, and Cloud Reliability - Finding out Why Cloud Security Matters - Establishing a Cloud Governance Strategy - Managing Service Levels - Summary - Self-Assessment Questions - Self-Study Topics: Identity and Access Management (IAM), Data Encryption
Chapter 14	Solution for Business Analytics Workloads - MapReduce - NoSQL - Summary - Self-Assessment Questions - Self-Study Topics: Data Warehousing, Business Intelligence (BI)

d. Self-study topics for Advance learners:

- Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid), Cloud Migration Strategies, Cloud Governance, High-Performance Computing

(HPC), Grid Computing, Distributed Computing, Service-Oriented Architecture (SOA), Hadoop Ecosystem, Big Data Analytics, Cluster Orchestration, Virtualization, Parallel I/O, Distributed File Systems, Computer-Aided Design (CAD), Computer-Aided Engineering (CAE), Bioinformatics, Drug Discovery, Risk Analytics, Algorithmic Trading, Serverless Computing, Containers, Service Level Agreements (SLAs), Cloud Monitoring Tools, Identity and Access Management (IAM), Data Encryption, Data Warehousing, Business Intelligence (BI).

e. Textbooks / Reference Books

1. Gautam Shroff, "Enterprise Cloud Computing: Technology, Architecture, Applications", Cambridge University Press, 2010.
2. Kai Hwang, Jack Dongarra, Geoffrey C. Fox, "Distributed and Cloud Computing: From Parallel Processing to the Internet of Things", Morgan Kaufmann, 2011.
3. Thomas Erl, Ricardo Puttini, Zaigham Mahmood, "Cloud Computing: Concepts, Technology & Architecture", Prentice Hall, 2013.
4. John Rhoton, Risto Haukioja, "Cloud Computing Architected: Solution Design Handbook", Recursive Press, 2011.

SQL FOR DATA ANALYTICS

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Subject: Major Project

Program Syllabus: Semester IV - Full Stack Development

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT754	Web Services - Rest API, ReactJS, NodeJS Development	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of web services, their architecture, and components.
2. To explore SOAP and RESTful web services, their advantages, and differences.
3. To learn about Java web services and REST API development using various tools and frameworks.
4. To gain hands-on experience in developing web applications using Node.js, MySQL, and MongoDB.

b. Course Outcomes

CO1	Understand the concepts, characteristics, and architecture of web services
CO2	Differentiate between SOAP and RESTful web services and their use cases.
CO3	Develop REST APIs using Java and implement advanced concepts like caching and versioning.
CO4	Build web applications using Node.js and integrate with databases like MySQL and MongoDB.
CO5	Apply the knowledge of web services and REST APIs to develop real-world applications.

c. Syllabus

Module -1	Introduction to Web Services and SOAP
Chapter 1	Web Services - Introduction to Web Services - Need for Web Services - Characteristics of Web Services - The architecture of Web Services - Components of Web Services - Security of Web Services - Standards - Summary - Self-Assessment Questions - Self-Study Topics: XML, JSON
Chapter 2	Web Services Resources - How do web services work? - Why Web Services? - Service Transport - Examples Of Web Services - Web Hosting Environment - Database and Management - Programming Technologies - Servers Functioning - Data Structures - Summary - Self-Assessment Questions - Self-Study Topics: HTTP, HTTPS

Chapter 3	SOAP Web Services - What Is SOAP? - Introduction to SOAP - Advantages & Disadvantages of SOAP Web Services - SOAP Building Blocks - SOAP Messaging Structure - SOAP Envelope Element - The Fault Message - SOAP Communication Mode - SOAP EXAMPLE - RESTful Web Services - Advantages of RESTful Web Services - SOAP vs REST - SOA - What Is Service? - Service Connections - Service-Oriented Terminologies - Characteristics of SOA - Components of service-oriented architecture - Functional aspects - Quality of Service aspects - Advantages of SOA - Summary - Self-Assessment Questions - Self-Study Topics: WSDL, UDDI
Chapter 4	Java Web Services - JAX Web Services - RPC Web Services - JAX Web Services Ex RPC - JAX Web Services Ex Document - Summary - Self-Assessment Questions - Self-Study Topics: Tomcat, Glassfish
Module -2	REST API and Database Integration

Chapter 5	Rest API - Introduction - The Need for Rest - REST API- Restful Web - Comparison of APIs - An Intuition Understanding of REST - REST Constraints - Concept of Serialization - Concept of Deserialization - Richardson Maturity Model - REST API- Tools - Summary - Self-Assessment Questions - Self-Study Topics: Postman, Swagger
Chapter 6	Request & Response - Understanding HTTP Request - HTTP Request Method - Analyze HTTP Response - Designing REST URLs - Summary - Self-Assessment Questions - Self-Study Topics: HTTP Status Codes, REST Best Practices
Chapter 7	RESTful Services - Controllers and Action - Creating Routing Templates - Understanding Routing Attributes - Using Parameters in Request - Model Variation - Summary - Self-Assessment Questions - Self-Study Topics: MVC Architecture, Spring Boot
Chapter 8	Database - REST API - Creating Domain Models - Scaffolding- REST API - Controllers - REST API - Database Seeding - REST API - Migration - REST API - Using DTOs - Implementing Paging - CORS AND Enabling CROS - Deferral Execution - Summary - Self-Assessment Questions - Self-Study Topics: ORM, JPA

Chapter 9	Caching - Introduction to Caching - REST API Expiration Model - Validation Model - REST API - Cache-Control - REST API - Concurrency in REST API - JSON - REST API - Cache Model - Summary - Self-Assessment Questions - Self-Study Topics: Redis, Memcached
Module -3	Advanced REST API and Node.js
Chapter 10	REST API - Advanced Concepts - Understanding HATEOAS - Approaches to Reterening Hypermedia Data - HAL and Collection + JSON - Versioning REST APIs - Summary - Self-Assessment Questions - Self-Study Topics: API Documentation, API Gateway
Chapter 11	Introduction to Node.js - What is Node.Js? - Features of Node Js - Who uses Node.Js? - Where to use Node.Js? - Node.Js Console - Node.Js RPEL - Node.Js NPM - Summary - Self-Assessment Questions - Self-Study Topics: Event-Driven Programming, Non-Blocking I/O

Chapter 12	Node.js - Node.js CL Options - Node.js Globals - Node.js OS - Node.js Errors - Node.js DNS - Node.js Debugger - Node.js Process - Node.js Child Process - Node.js Buffers - Node.js Streams - Node.js File System - Node.js Path - Node.js StringDecoder - Node.js Query String - Node.js Assertion - Node.js Callbacks - Node.js Events - Node.js Web Modules - Summary - Self-Assessment Questions - Self-Study Topics: Express.js, Socket.I
Chapter 13	Node.js MySQL - MySQL Create Connection - MySQL Create Database - MySQL Create Table - MySQL Insert Record - MySQL Update Record - MySQL Delete Record - MySQL Select Record - MySQL Select Unique - MySQL Drop Table - Summary - Self-Assessment Questions - Self-Study Topics: Sequelize ORM, MySQL Workbench

Chapter 14	Node.js MongoDB - Create Connection - Create Database - Create Collection - MongoDB Insert - MongoDB Select - MongoDB Query - MongoDB Sorting - MongoDB Remove - Summary - Self-Assessment Questions - Self-Study Topics: Mongoose ODM, MongoDB Atlas
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d. Self-study topics for Advance learners:

- XML, JSON, HTTP, HTTPS, WSDL, UDDI, Tomcat,
- Glassfish, Postman, Swagger, HTTP Status Codes,
- REST Best Practices, MVC Architecture, Spring Boot,
- ORM, JPA, Redis, Memcached, API Documentation,
- API Gateway, Event-Driven Programming,
- Non-Blocking I/O, Express.js, Socket.IO, Sequelize ORM,
- MySQL Workbench, Mongoose ODM, MongoDB Atlas.

e. Textbooks / Reference Books

1. Leonard Richardson, Sam Ruby, "RESTful Web Services", O'Reilly Media, 2007.
2. Ethan Cerami, "Web Services Essentials: Distributed Applications with XML-RPC, SOAP, UDDI & WSDL", O'Reilly Media, 2002.
3. Mario Casciaro, Luciano Mammino, "Node.js Design Patterns: Master best practices to build modular and scalable server-side web applications", Packt Publishing, 2016.
4. Ian Robinson, Jim Webber, Emil Eifrem, "Graph Databases: New Opportunities for Connected Data", O'Reilly Media, 2015.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT755	DevOps - 2 (Ansible, Puppet, Nagios)	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of DevOps, its history, goals, and stakeholders.
2. To explore various SDLC models, software testing methodologies, and Agile practices.
3. To gain hands-on experience with configuration management tools like Ansible and Puppet.
4. To learn about continuous monitoring using Nagios and its installation and configuration.

b. Course Outcomes

CO1	Understand the concepts, principles, and ecosystem of DevOps.
CO2	Apply Agile methodologies and software testing techniques in the DevOps lifecycle.
CO3	Utilize Ansible for provisioning, configuration management, and automation.
CO4	Implement configuration management using Puppet and its language basics.
CO5	Set up and configure Nagios for continuous monitoring of hosts and services.

c. Syllabus

Module -1	Introduction to DevOps and Agile Methodologies
Chapter 1	Introduction to DevOps - Introduction to DevOps - What is DevOps? - SDLC models, Lean, ITIL, Agile - Why DevOps? - History of DevOps - DevOps Stakeholders - DevOps Goals - Important terminology - DevOps perspective - DevOps and Agile - DevOps Tools - Configuration management - Continuous Integration and Deployment - Self-Study Topics: DevOps Culture, DevOps Maturity Model
Chapter 2	Overview of DevOps - Why DevOps? - DevOps Market Trends - DevOps Engineer Skills - DevOps Delivery Pipeline - DevOps Ecosystem - Self-Study Topics: DevOps and Cloud Computing, DevOps and Microservices
Chapter 3	Introduction to SDLC, Software testing, Agile: Software testing lifecycle - Working with Block box testing - Working with White box testing - Working Grey box testing - Working with Function testing - Working with Regression testing, smoke testing, System testing, Integration testing etc. - Self-Study Topics: Test-Driven Development (TDD), Behavior-Driven Development (BDD)
Chapter 4	Agile Methodologies - Process flow of Scrum Methodologies - Project planning, scrum testing, sprint Planning and Release management - Analysis - Design, Execution and wrapping closure - Self-Study Topics: Kanban, Lean Software Development
Module -2	Configuration Management with Ansible and Puppet

Chapter 5	Introduction to Ansible - Introduction to Ansible - What is Ansible? - Change Management - Provisioning with Ansible - Benefits of using Ansible - Self-Study Topics: Ansible Tower, Ansible Galaxy
Chapter 6	Ansible Building blocks and Process flow - Introduction to Ansible Anatomy - Ansible Requirements Specification - Overview of Ansible Components - Overview of Ansible Strategy - Self-Study Topics: Ansible Vault, Ansible Roles
Chapter 7	Variable, Facts and jinja2 templates - Working with Ansible Variable - Working with Facts - Working with Jinja2 Template - Self-Study Topics: Ansible Filters, Ansible Plugins
Chapter 8	Ansible Playbook Modules and directory structure - Introduction to Ansible Playbook - Introduction to Ansible Modules - Playbook Language Example - Working on Ansible Handlers - Executing a Playbook - Self-Study Topics: Ansible Best Practices, Ansible and Docker
Chapter 9	Introduction to Puppet - Introduction to Puppet - What is Puppet? - Puppet Installation - Puppet Master and Agent Setup - Puppet Module - Node Classification - Puppet Environment - Puppet Classes - Automation & Reporting.co © 2018 All rights Reserved - Self-Study Topics: Puppet Enterprise, Puppet Bolt
Chapter 10	Puppet for configuration management - How puppet works - Puppet Architecture - Master and Agents - Puppet terminology and about Manifests - Puppet configuration files - Self-Study Topics: Puppet and Git, Puppet and Hiera

Chapter 11	Puppet Language Basics - The declarative language - Resources - Using Basic resources like file, exec, package service - Resource Collectors - Virtual Resources - Exported Resources - Manifests - Modules and Classes - Class Parameters - Defined Type - Self-Study Topics: Puppet DSL, Puppet and Ruby
Chapter 12	Puppet Forge - Understanding the Puppet Forge - Module structure - Install LAMP with pre-existing modules - Installing Apache Tomcat with Puppet Modules - Self-Study Topics: Writing Custom Puppet Modules, Puppet and AWS
Module -3	Continuous Monitoring with Nagios
Chapter 13	Introduction to Nagios - What is Nagios - Why Nagios? - Introduction to Continuous Monitoring - Nagios Architecture - Nagios commands - Nagios features - Nagios applications - Benefits of Nagios - Self-Study Topics: Nagios Plugins, Nagios and Grafana
Chapter 14	Nagios Installation and Configuration - Nagios Installation - Nagios Configuration - Nagios hosts and services - Self-Study Topics: Nagios and SNMP, Nagios and Prometheus

d. Self-study topics for Advance learners:

- DevOps Culture, DevOps Maturity Model,
- DevOps and Cloud Computing, DevOps and Microservices,
- Test-Driven Development (TDD), Behavior-Driven Development (BDD),
- Kanban, Lean Software Development,
- Ansible Tower, Ansible Galaxy, Ansible Vault, Ansible Roles, Ansible Filters, Ansible Plugins, Ansible Best Practices, Ansible and Docker,
- Puppet Enterprise, Puppet Bolt, Puppet and Git, Puppet and Hier,

- Puppet DSL, Puppet and Ruby, Writing Custom Puppet Modules,
- Puppet and AWS, Nagios Plugins, Nagios and Grafana,
- Nagios and SNMP, Nagios and Prometheus.

e. Textbooks / Reference Books

1. Gene Kim, Jez Humble, Patrick Debois, John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations", IT Revolution Press, 2016.
2. Lorin Hochstein, Rene Moser, "Ansible: Up and Running", O'Reilly Media, 2017.
3. James Turnbull, "Puppet 5 Beginner's Guide", Packt Publishing, 2017.
4. Wolfgang Barth, "Nagios: System and Network Monitoring", No Starch Press, 2008.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Subject: Major Project

Program Syllabus: Semester IV - Data Analytics

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT756	Data Analytics using R	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To introduce the fundamental concepts of data analytics and R programming language.
2. To develop skills for data manipulation, exploration, and visualization using R.
3. To learn various data mining techniques and machine learning algorithms using R.
4. To apply data analytics techniques for solving real-world problems.

b. Course Outcomes

CO1	Understand the concepts of data analytics and its applications.
CO2	Develop programs using R programming language for data analysis tasks.
CO3	Perform data manipulation, exploration, and visualization using R.
CO4	Apply data mining techniques and machine learning algorithms for predictive modeling
CO5	Analyze and interpret data to derive meaningful insights for decision-making

c. Syllabus

Module -1	Introduction to Data Analytics and R Programming
Chapter 1	Introduction to Data Analytics - What is Data Analytics? -What are Data Analytics Tools? -Evolution of Data Analytic Approaches - Data Analysis vs Data Reporting - Data Analysis Process - Types of Data Analysis - Characteristics of Data Analysis - Applications of Data Analysis - Skills required to become a Data Analyst - Introduction of Big Data Analytics - Technical & Business Skills for Data Analytic - Summary - Self-Assessment Questions SELF STUDY TOPIC Introduction to Business Analytics
Chapter 2	Introduction to R Programming - What is R Programming Language? - History of R - Why Learn R Programming Language - Features of R Programming - How R is better than Other Technologies - R Scripts - Applications of R Programming - Advantages of R Programming language - Summary - Self-Assessment Questions SELF STUDY TOPIC R Development Environment
Chapter 3	Basic Concepts of R - How to Install R - How to Install R Studio - Basic R Concepts - Basic Interaction with R Console - Keywords - Data Types - Data Structures - Summary - Self-Assessment Questions SELF STUDY TOPIC R Operators

Chapter 4	R Packages - Packages in R - How to Install R Packages for Windows - How to Install R Packages for Linux - How to Use Packages in R - List of Packages in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Creating Custom R Packages
Module -2	Data Manipulation and Import Techniques
Chapter 5	Control Structures and Functions using R - Control Structures in R Programming - Functions in R - Scope of R Function - What are R Vector Functions - R Numeric Functions - Character Functions in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Recursive Functions in R
Chapter 6	Data Manipulation in R - What is Data Manipulation in R? - Creating Subsets of Data in R - sample() command in R - Adding Calculated Fields to Data - Creating Subgroups or Bins of Data - Combining and Merging Datasets in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Handling Missing Data in R
Chapter 7	Data Import Techniques in R - Process of Importing Data in R - Packages installation used for database import - Connect to RDBMS from R using ODBC and basic SQL queries in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Importing Data from NoSQL Databases
Chapter 8	Exploratory Data Analysis - Introduction to Exploratory Data Analysis - Summary Statistics - Data Visualization Techniques - Summary - Self-Assessment Questions SELF STUDY TOPIC Advanced Exploratory Data Analysis Techniques

Module -3	Data Visualization and Data Mining
Chapter 9	Data Visualization in R - What is R Data Visualization? - R Visualization Packages - Use of R Programming - R Graphics - Data Visualization in R using ggplot2 - What to Learn in Data Visualization in R? - Advantages of Data Visualization in R - Disadvantages of Data Visualization in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Interactive Data Visualization in R
Chapter 10	Data Mining: Clustering Techniques - Introduction to Data Mining - Understanding Machine Learning - Supervised and Unsupervised Machine Learning Algorithms - K-means Clustering - Summary - Self-Assessment Questions SELF STUDY TOPIC Hierarchical Clustering Techniques
Chapter 11	Data Mining: Association Rule Mining & Collaborative Filtering - Association Rule Mining - User Based Collaborative Filtering (UBCF) - Item Based Collaborative Filtering (IBCF) - Summary - Self-Assessment Questions SELF STUDY TOPIC Advanced Association Rule Mining Techniques
Chapter 12	Linear and Logistic Regression - Linear Regression - Logistic Regression - Summary - Self-Assessment Questions SELF STUDY TOPIC Regularization Techniques in Regression
Module -4	Advanced Data Mining Technique
Chapter 13	Anova and Sentiment Analysis - Anova - Sentiment Analysis - Summary - Self-Assessment Questions SELF STUDY TOPIC Advanced Sentiment Analysis Techniques

Chapter 14	Data Mining: Decision Trees and Random Forest - What is R Decision Trees? - Applications of Decision Trees - How to Create Decision Trees in R - Common R Decision Trees Algorithms - Guidelines for Building Decision Trees in R - Decision Tree Options - Advantages of R Decision Trees - Disadvantages of R Decision Trees - Introduction to Random Forest in R - Summary - Self-Assessment Questions SELF STUDY TOPIC Ensemble Learning Techniques
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d. Self-study topics for Advance learners:

- Introduction to Business Analytics
- R Development Environment
- R Operators
- Creating Custom R Packages
- Recursive Functions in R
- Handling Missing Data in R
- Importing Data from NoSQL Databases
- Advanced Exploratory Data Analysis Techniques
- Interactive Data Visualization in R
- Hierarchical Clustering Techniques
- Advanced Association Rule Mining Techniques
- Regularization Techniques in Regression
- Advanced Sentiment Analysis Techniques
- Ensemble Learning Techniques

e. Textbooks / Reference Books

1. Garrett Golemund, Hadley Wickham, "R for Data Science," O'Reilly Media, Inc., 2017.
2. Jared P. Lander, "R for Everyone: Advanced Analytics and Graphics," Addison-Wesley Professional, 2017.
3. Paul Teetor, "R Cookbook: Proven Recipes for Data Analysis, Statistics, and Graphics," O'Reilly Media, Inc., 2011.
4. Max Kuhn, Kjell Johnson, "Applied Predictive Modeling," Springer Science & Business Media, 2013.
5. Chandan K. Reddy, "Data Analytics Using R: Principles and Examples," Packt Publishing Ltd, 2022.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT757	Data Analytics for Decision Making	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of data analytics and its importance in decision-making processes.
2. To learn and apply Python programming for data analysis and manipulation using libraries such as NumPy and Pandas.
3. To gain knowledge of statistical concepts, probability, and machine learning techniques for data analysis.
4. To develop skills in data wrangling, visualization, and creating interactive dashboards using tools like Matplotlib, Seaborn, and Tableau.

b. Course Outcomes

CO1	Understand the concepts, types, and importance of data analytics in decision-making processes.
CO2	Apply Python programming and SQL for data manipulation, analysis, and database management.
CO3	Utilize statistical concepts, probability, and machine learning techniques for data analysis and modeling.
CO4	Perform data wrangling and visualization using Python libraries like Matplotlib and Seaborn.
CO5	Create interactive dashboards and visualizations using Tableau for effective data communication and decision-making.

c. Syllabus

Module -1	Introduction to Data Analytics and Python
Chapter 1	Introduction to Data Analytics - Introduction - What is data analytics? - Types of Data Analytics - Importance of Data Analytics - Tools used in Data Analytics - Role of Data Analyst - Summary - Self-Assessment Question - References/Reference Reading - Self-Study Topics: Data Analytics Lifecycle, Data Analytics Ethics
Chapter 2	- Introduction - History - Application of Python - Python 2 vs. Python 3 - Python Installation (Environment Set-up) - Installation on Windows - Installation on Mac - First Python Program - Basic Syntax of Python - Python data types - Numbers - Sequence Type - Dictionary - Boolean - Python Operators - Python If-else statements - Python Loops - Python Function - Summary - Self-Assessment Questions - References - Self-Study Topics: Python Libraries, Python Object-Oriented Programming

Chapter 3	NumPy & Pandas - Introduction - History - The NumPy Installation - Narray: The Core of the Library - Create an Array - Basic Operations - Arithmetic Operators - The Matrix Product - Slicing - Pandas - Installation of pandas - Why Use Pandas? - Pandas Data Structures - The Series - The DataFrame - Summary - Self-Assessment learning - References - Self-Study Topics: NumPy Broadcasting, Pandas Data Manipulation
Chapter 4	Getting Started with SQL - Introduction - What's a database? - Types of SQL Statements - Creating Database - CREATE Clause - SQL ALTER (ADD, DROP, MODIFY) - SQL SELECT, ORDER BY, DISTINCT - SQL - LIKE, BETWEEN, IN - Summary - Self-assessment learning - References - Self-Study Topics: SQL Joins, Subqueries, and Views
Module -2	Data Analysis Techniques and Machine Learning

Chapter 5	Introduction to Data-set - Introduction to data Set - Characteristics of Data Sets - Qualitative and Quantitative Data - Types of Data Set's - Record Data - Graph-based Data - Ordered Data - Training Dataset - Validation Dataset - Test Dataset - Dataset Split Ratio - Summary - Self-Assessment learning - References - Self-Study Topics: Data Preprocessing, Feature Engineering
Chapter 6	Descriptive and Inferential Statistics - Introduction - Descriptive Statistics - Inferential Statistics - Basic Terminologies - Types of Data - Categorical Data - Numerical Data - Why Data Types are important? - Mean, Median, Mode, and Range Definitions - Summary - Self-Assessment Learning - References - Self-Study Topics: Measures of Dispersion, Correlation and Regression
Chapter 7	Probability - Introduction - Probability Space - Random Variable - Discrete Random Variable - Continuous Random Variable - The Probability Rules - What Is Expectation? - Variance and Covariance - Probability Distributions - Summary - Self-Assessment Questions - References - Self-Study Topics: Joint Probability, Conditional Probability

Chapter 8	Discrete Probability - Introduction - Continuous probability distribution - The Normal Probability Distribution - Discrete Probability Distributions - Binomial Distribution - Criteria - Central Limit Theorem - Bayes' Theorem - Hypothesis Testing - Statistical Hypotheses - Decision Errors - Decision Rules - Confidence Intervals - Summary - Self-Assessment Learning - References - Self-Study Topics: Poisson Distribution, Chi-Square Distribution
Chapter 9	Supervised Learning - Introduction - Types of Supervised Learning - Logistic regression - Decision Tree - Naive Bayes algorithm - KNN - Support Vector Machines - Summary - Self-Learning Assessment - References - Self-Study Topics: Random Forests, Gradient Boosting
Chapter 10	Regression - Introduction - Terminologies Related to the Regression Analysis - Regression Analysis - Linear Regression - Multiple Regression - Polynomial Regression - Summary - Self-Learning Assessment - References - Self-Study Topics: Ridge Regression, Lasso Regression

Chapter 11	Unsupervised learning - Introduction - Clustering - Clustering Algorithms - K-Means Clustering Algorithm - Hierarchical Clustering - Summary - Self – Learning Assessment - References - Self-Study Topics: DBSCAN, Principal Component Analysis (PCA)
Module -3	Data Wrangling and Visualization
Chapter 12	- Introduction - Definition of Data Wrangling - Data Wrangling Vs Data Mining - Need of Data Wrangling - Data Wrangling Use-Cases - Data Wrangling Benefits - Data Wrangling Challenges - Tools and techniques of data wrangling - Role of machine learning in data wrangling - Summary - Self-Assessment Learning - Reference - Self-Study Topics: Data Cleaning, Data Transformation
Chapter 13	Python Visualization - Introduction - Data Visualisation - Matplotlib - Anatomy of Matplotlib Figure - Histogram - Pie Chart - Seaborn - Strip plot - Swarmplot - Barplot - Countplot - Summary - Self-Learning Assessment - References - Self-Study Topics: Heatmaps, Subplots

Chapter 14	Premium Visualization by Tableau - Introduction - Tableau Software - Why Tableau - Tableau Product Suite - Download and Install Tableau Public - Working of Tableau - Spreadsheet v/s Data Visualization - Summary - Self-Learning Assessment - References - Self-Study Topics: Tableau Dashboards, Tableau Stories
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d. Self-study topics for Advance learners:

- Data Analytics Lifecycle
- Data Analytics Ethics
- Python Libraries
- Python Object-Oriented Programming
- NumPy Broadcasting
- Pandas Data Manipulation
- SQL Joins, Subqueries, and Views
- Data Preprocessing
- Feature Engineering
- Measures of Dispersion
- Correlation and Regression
- Joint Probability, Conditional Probability
- Poisson Distribution, Chi-Square Distribution
- Random Forests, Gradient Boosting
- Ridge Regression, Lasso Regression
- DBSCAN, Principal Component Analysis (PCA)
- Data Cleaning, Data Transformation
- Heatmaps, Subplots, Tableau Dashboards, Tableau Stories.

e. Textbooks / Reference Books

1. Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", Morgan Kaufmann, 2011.
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Springer, 2009.
3. Wes McKinney, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython", O'Reilly Media, 2017.
4. Alberto Cairo, "The Truthful Art: Data, Charts, and Maps for Communication", New Riders, 2016.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Subject: Major Project

Program Syllabus: Semester IV - Artificial Intelligence and Machine Learning

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT758	Big Data Hadoop	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of big data and its challenges in handling structured, semi-structured, and unstructured data.
2. To explore the Apache Hadoop ecosystem and its components for big data processing and storage.
3. To learn and implement the MapReduce programming model for processing large datasets.
4. To gain hands-on experience with Pig, Hive, and HBase for data processing, querying, and storage in a Hadoop environment.

b. Course Outcomes

CO1	Understand the basics of big data, its challenges, and real-world applications.
CO2	Comprehend the architecture and components of the Apache Hadoop ecosystem for big data processing and storage.
CO3	Develop and execute MapReduce programs for processing large datasets in a distributed computing environment.
CO4	Utilize Pig and Hive for data processing and querying in a Hadoop environment.
CO5	Implement HBase for real-time read/write access to large datasets in a Hadoop ecosystem.

c. Syllabus

Module -1	Big Data Basics and Applications
Chapter 1	Big Data Basics - Big data Introduction - Human generated data Vs Machine generated data - Challenges of big data (Structured, Semi structured, Un-structured data) - Technologies for Big data support - Examples of Big data in the real world - Computing systems - Big data open-source solution – Apache Hadoop - How Hadoop solves Big data problem - RDBMS Vs Hadoop - Big data terms - Summary - Self-Assessment Questions - Self-Study Topics: 4 V's of Big Data, Lambda Architecture
Chapter 2	Big data Applications - Introduction - Big data analytics technologies and tools - Marketing applications - Fraud Detections applications - Risk analysis applications - Health care applications - Summary - Self-Assessment Questions - Self-Study Topics: Social Media Analytics, IoT Analytics
Module -2	Hadoop Distributed File System (HDFS) and MapReduce

Chapter 3	Hadoop Distributed File System (HDFS) - Introduction to Apache Hadoop - Apache Hadoop ecosystem - Hadoop Vs Java - HDFS: Hadoop Distributed File System - Features of HDFS - Daemons of Hadoop - Replication and data organization in HDFS - Accessing HDFS - Unix commands - Read/Write steps in HDFS - Installation and Set-up of Hadoop - HDFS Access - Hadoop serialization programs - HDFS read/write architecture - Summary - Self-Assessment Questions - Self-Study Topics: HDFS Federation, HDFS High Availability
Chapter 4	Map Reduce Programming Model - Map Reduce introduction - Applications of Map Reduce - Map Reduce Process steps - Map Reduce Architecture - Map Reduce working Examples - Built in MapReduce Algorithms - Writing MapReduce Programs - Summary - Self-Assessment Questions - Self-Study Topics: Combiner, Partitioner
Module -3	Pig, Hive, and HBase
Chapter 5	PIG - PIG Introduction - Map Reduce Vs. Apache Pig - SQL vs. Apache Pig - Installation of PIG - Different data types in Pig - Modes of Execution in Pig - Loading data - Exploring Pig Latin commands - Sample PIG programs - Self-Study Topics: User-Defined Functions (UDFs) in Pig, Pig Streaming

Chapter 6	HIVE - Hive introduction - Hive architecture - Hive Vs RDBMS - Installation of HIVE - HiveQL and the shell - Managing tables (external Vs managed) - Data types and schemas - Partitions and buckets - HIVE commands - Sample HIVE programs - Summary - Self-Assessment Questions - Self-Study Topics: Hive Indexes, Hive Transactions
Chapter 7	HBASE - Introduction - Hbase Architecture - HBase vs. RDBMS - Hbase Installation - Hbase commands - Read/write steps in Hbase - Sample HBASE program - Summary - Self-Assessment Questions - Self-Study Topics: HBase Coprocessors, HBase Bulk Loading

d. Self-study topics for Advance learners:

4 V's of Big Data, Lambda Architecture, Social Media Analytics, IoT Analytics, HDFS Federation, HDFS High Availability, Combiner, Partitioner, User-Defined Functions (UDFs) in Pig, Pig Streaming, Hive Indexes, Hive Transactions, HBase Coprocessors, HBase Bulk Loading.

e. Textbooks / Reference Books

1. Tom White, "Hadoop: The Definitive Guide", O'Reilly Media, 2015.
2. Alan Gates, "Programming Pig: Dataflow Scripting with Hadoop", O'Reilly Media, 2016.
3. Edward Capriolo, Dean Wampler, Jason Rutherglen, "Programming Hive: Data Warehouse and Query Language for Hadoop", O'Reilly Media, 2012.
4. Lars George, "HBase: The Definitive Guide", O'Reilly Media, 2011.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Program Syllabus:

S. N	Course Code	Course Title	Course Type	Credit	Week
1	24ONMCT759	IOT Cloud and Watson Analytics	Prog. Core	4	12
PRE-REQUISITE		-			

a. Course Objectives

1. To understand the fundamentals of networking, IoT, and its architecture.
2. To explore the key technologies and mechanisms in IoT, including identification, device intelligence, and sensor technology.
3. To learn about the integration of cloud computing and IoT, and the use of Watson IoT Platform and analytics.
4. To comprehend the resource management, privacy, security, and governance aspects of IoT.
5. To study various IoT applications and real-world case studies across different domains.

b. Course Outcomes

CO1	Understand the basics of networking, IoT, its history, advantages, and architectural frameworks.
CO2	Identify and apply the fundamental IoT mechanisms and key technologies in real-world scenarios.
CO3	Integrate cloud computing with IoT and utilize Watson IoT Platform and analytics for data management and cognitive computing.
CO4	Implement effective resource management techniques and address privacy, security, and governance challenges in IoT.
CO5	Analyze and develop IoT applications across various domains, such as healthcare, smart cities, and agriculture, based on case studies.

c. Syllabus

Module -1	Introduction to IoT and Key Technologies
Chapter 1	Introduction to IoT - Introduction to Networking - Wired and Wireless Networks - Devices used in Networking - Introduction to IoT - History of IoT - Advantages of IoT - Overview - Characteristics of IoT - IoT Definitions - IoT Detailed Architecture - Three and five layer architectures - Cloud and Fog Based Architectures - Representative Architecture - Service Oriented architecture (SOA) - API Oriented Architecture - IoT Framework - Big Data Analytics - Smart Objects - Smart Applications - Self-Study Topics: Edge Computing, IoT Communication Protocols
Chapter 2	Fundamental IOT Mechanism and Key Technologies - Internet Principles - Identification of IoT Objects and Services - Device Intelligence - Mobility Support - Device Power - Sensor Technology - RFID Technology - Satellite Technology - Raspberry Pi & Arduino device - Self-Study Topics: NFC Technology, LoRaWAN
Module -2	IoT Cloud, Watson Analytics, and Resource Management

Chapter 3	IoT Cloud and Watson Analytics - Introduction to cloud computing - The internet of Things and Cloud Computing - Mobile Cloud Computing - Integration of Cloud computing and Internet of Things(CloudIoT) - Main CloudIoT drivers - Internet of Things to Smart IoT Through Semantic, Cognitive, and Perceptual Computing - Cognitive computing and the Internet of Things - Introduction to Watson IoT Platform - Important concepts in the Watson IoT Platform - Watson IoT Platform Feature Overview - Standards and requirements - IBM Cloud and the Watson IoT Platform - Introduction to data management - Introduction to cognitive computing - Introduction to Watson Analytics - Watson APIs: Build with Watson - IBM Watson applied to industries, businesses, and science - Watson use cases - Self-Study Topics: AWS IoT, Azure IoT
Chapter 4	Resource Management in Internet of Things - Clustering - Software Agents - Clustering Principles in an Internet of Things Architecture - Design Guidelines - Software Agents for Object Representation - Data Synchronization - Identity Portrayal - Identity Management - Federated Identity Management Model - User-Centric Identity Management - Device Centric Identity Management and Hybrid-Identity Management - Self-Study Topics: Distributed Resource Management, IoT Energy Management
Module 3	IoT Security, Privacy, Governance, and Applications

Chapter 5	Internet of Things: Privacy, Security and Governance - Issues and Challenges in IoT Security - Vulnerabilities of IoT - Security Requirements - Threat Analysis - Use Cases and misuse cases - Activity Modeling and Threats - Security Threats at different layers of IoT Architecture - Identity Establishment - Access control - Non-Repudiation and Availability - Security model for IoT - Providing security on layers to defend IoT - Security Measures for IoT Platforms/Devices - IoT Governance - Self-Study Topics: IoT Privacy Regulations, Blockchain for IoT Security
Chapter 6	IOT Applications and Case Studies - Connected Life - Consumer and socio-economic impact - Distinctive features of IOT - Application-Cloud with IoT in HealthCare - Application- Smart cities and communities - Application- Smart home and smart metering - Application- Automotive and Smart mobility - IoT Some Examples - Patras: Internet of Things Case Study - AirQ: Air Quality Internet of Things Case Study - Case Study: Transforming to an engaged and connected city(The City of Mississauga) - Internet of Things in Agriculture: a Case Study of Smart Dairy Farming in Ontario, Canada - HP Turns Printers into IoT Hubs in the Home - Self-Study Topics: Industrial IoT (IIoT), IoT in Supply Chain Management

d. Self-study topics for Advance learners:

Edge Computing, IoT Communication Protocols, NFC Technology, LoRaWAN, AWS IoT, Azure IoT, Distributed Resource Management, IoT Energy Management, IoT Privacy Regulations, Blockchain for IoT Security, Industrial IoT (IIoT), IoT in Supply Chain Management.

e. Textbooks / Reference Books

1. Rajkumar Buyya, Amir Vahid Dastjerdi, "Internet of Things: Principles and Paradigms", Morgan Kaufmann, 2016.

2. Arshdeep Bahga, Vijay Madisetti, "Internet of Things: A Hands-On Approach", VPT, 2014.
3. Honbo Zhou, "The Internet of Things in the Cloud: A Middleware Perspective", CRC Press, 2012.
4. Perry Lea, "Internet of Things for Architects: Architecting IoT solutions by implementing sensors, communication infrastructure, edge computing, analytics, and security", Packt Publishing, 2018.

f. Assessment Pattern

Internal Assessment Weightage (%)	External Assessment Weightage (%)	Total Weightage(%)
30	70	100

Subject: Major Project